

- Q1** The number of positive integers not exceeding 100 that are either odd or the square of an integer is ____
 (A) 63 (B) 59
 (C) 55 (D) 50
- Q2** How many ways are there to pack six copies of the same book into four identical boxes, where a box contains six books?
 (A) 4 (B) 6
 (C) 7 (D) 9
- Q3** Which of the following pairs of propositions are not logically equivalent?
 (A) $((p \rightarrow r) \wedge (q \rightarrow r))$ and $((p \vee q) \rightarrow r)$
 (B) $p \leftrightarrow q$ and $(\neg p \leftrightarrow \neg q)$
 (C) $((p \wedge q) \vee (\neg p \wedge \neg q))$ and $p \leftrightarrow q$
 (D) $((p \wedge q) \rightarrow r)$ and $((p \rightarrow r) \wedge (q \rightarrow r))$
- Q4** Consider the following linear programming (LP):
 Max. $z = 2x_1 + 3x_2$
 Such that $2x_1 + x_2 \leq 4$
 $x_1 + 2x_2 \leq 5$
 $x_1, x_2 \geq 0$
 The optimum value of the LP is
 (A) 23 (B) 9.5
 (C) 13 (D) 8
- Q5** Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines used with this machine. How many bits will be there in Tag, line and word field of format of main memory addresses?
 (A) 8, 5, 3 (B) 8, 6, 2
 (C) 7, 5, 4 (D) 7, 6, 3
- Q6** The following program is stored in memory unit of the basic computer. What is the content of the accumulator after the execution of program? (All location numbers listed below are in hexadecimal).

Location Instruction

210 CLA
 211 ADD 217
 212 INC
 213 STA 217
 214 LDA 218
 215 CMA
 216 AND 217
 217 1234H
 218 9CE2H

(A) 1002H
 (B) 2011H
 (C) 2022H
 (D) 0215H

- Q7** A non-pipeline system takes 50 ns to process a task. The same task can be processed in six-segment pipeline with a clockcycle of 10ns. Determine approximately the speedup ratio of the pipeline for 500 tasks.
 (A) 6 (B) 4.95
 (C) 5.7 (D) 5.5
- Q8** What is the radix of the numbers if the solution to the quadratic equation $x^2 - 10x + 26 = 0$ is $x = 4$ and $x = 7$?
 (A) 8 (B) 9
 (C) 10 (D) 11
- Q9** Consider the following pseudo-code fragment, where a and b are integer variables that have been initialized:
 /* Pre-conditions : $(a > 1 \wedge a < b)$ */
 /* Assume that overflow never occurs */
 int $x = 0$; int $p = 1$;
 while ($p < b$) {
 $p = p * a$;
 $x = x + 1$;
 }
 When the while loop terminates, what will be the value of x in terms of a and b ?



- (A) a^b
 (B) b^a
 (C) $\lfloor \log_a^b \rfloor / * \lfloor \rfloor$ means floor $*/$
 (D) $\lfloor \log_a^b \rfloor / * \lceil \rceil$ means ceil $*/$

- Q10** In HTML, `< map >` tag is used for
 (A) defining a path between two nodes in an image
 (B) defining clickable region in an image
 (C) highlighting an area in an image
 (D) defining the site-map of a web-site

- Q11** Suppose you are compiling on a machine with 1-byte chars, 2-byte shorts, 4-byte ints, and 8-byte doubles, and with alignment rules that require the address of every primitive data element to be an integer multiple of the element's size. Suppose further that the compiler is not permitted to reorder fields; padding is used to ensure alignment. How much space will be consumed by the following array?

```
struct {
    short s;
    char c;
    short t;
    char d;
    double r;
    int i;
} A[10]; /*10-element array of structs*/
```

- (A) 150 bytes (B) 320 bytes
 (C) 240 bytes (D) 200 bytes

- Q12** Consider the following recursive Java function f that takes two long arguments and returns a float value:

...

```
public static float f(long m, long n) {
    float result = (float)m / (float)n;
    if (m < 0 || n < 0) return 0.0 f;
    else result -= f(m * 2, n * 3);
    return result;
}
```

...

Which of the following real values best approximates the value of $f(1, 3)$?

- (A) 0.2 (B) 0.4

- (C) 0.6 (D) 0.8

- Q13** The data node and name node in HADOOP are
 (A) Both Worker Nodes
 (B) Both Master Nodes
 (C) Worker Node and Master Node respectively
 (D) Master Node and Worker Node respectively

- Q14** Consider a relational schema $S = (U, V, W, X, Y, Z)$ on which the following functional dependencies hold:
 $\{U \rightarrow V, VW \rightarrow X, Y \rightarrow W, X \rightarrow U\}$
 Which are the candidate keys among following options?
 (A) UY, VY (B) UY, VY, XY
 (C) UYZ, VYZ, VWZ (D) UYZ, VYZ, XYZ

- Q15** Consider a hypothetical machine with 3 pages of physical memory, 5 pages of virtual memory, and $\langle A, B, C, D, A, B, E, A, B, C, D, E, B, A, \text{ as } B \rangle$ the stream of page references by an application. If P and Q are the number of page faults that the application would incur with FIFO and LRU page replacement algorithms respectively, then $(P, Q) = \underline{\hspace{1cm}}$. (Assuming enough space for storing 3 page frames)
 (A) (11, 10) (B) (12, 11)
 (C) (10, 11) (D) (11, 12)

- Q16** Consider a disk system having 60 cylinders. Disk requests are received by a disk drive for cylinders 10, 22, 20, 2, 40, 6 and 38, in that order. Assuming the disk head is currently at cylinder 20, what is the time taken to satisfy all the requests if it takes 2 milliseconds to move from one cylinder to adjacent one and Shortest Seek Time First (SSTF) algorithm is used?
 (A) 240 milliseconds
 (B) 96 milliseconds
 (C) 120 milliseconds
 (D) 112 milliseconds

- Q17** Suppose you have a Linux file system where the block size is 2 K bytes, a disk address is 32 bits, and an i-node contains the disk addresses of the first 12 direct blocks of file, a single indirect



block, and a double indirect block.

Approximately, what is the largest file that can be represented by an i-node?

- (A) 513 Kbytes (B) 513 MBytes
(C) 537 Mbytes (D) 537 KBytes

Q18 Consider a single-level page table system, with the page table stored in the memory. If the hit rate to TLB is 80%, and it takes 15 nanoseconds to search the TLB, and 150 nanoseconds to access the main memory, then what is the effective memory access time, in nanoseconds?

- (A) 185 (B) 195
(C) 205 (D) 175

Q19 Which of the following UML diagrams has a static view?

- (A) Collaboration diagram
(B) Use-Case diagram
(C) State chart diagram
(D) Activity diagram

Q20 Modifying the software by restructuring is called

- (A) Adaptive maintenance
(B) Corrective maintenance
(C) Perfective maintenance
(D) Preventive maintenance

Q21 A Company has a choice of two languages L_1 and L_2 to develop a software for their client. Number of LOC required to develop an application in L_2 is thrice the LOC in language L_1 . Also, software has to be maintained for next 10 years. Various parameters for two languages are given below to decide which language should be preferred for development.

PARAMETER	L_1	L_2
Man-year needed for development	LOC/1000	LOC/1000
Development cost	Rs. 70,000	Rs. 90,000
Cost of Maintenance per year	Rs. 1,00,000	Rs. 40,000

Total cost of project include cost of development and maintenance. What is the LOC for L_1 for

which cost of developing the software with both languages must be same?

- (A) 2000 (B) 6000
(C) 3000 (D) 5000

Q22 A Software project was estimated at 864 Function Points. A six person team will be assigned to project consisting of a requirement gathering person, one designer, two programmers and two testers. The salary of the designer is ₹ 70,000 per month, requirement gatherer is ₹ 50,000 per month, programmer is ₹ 60,000 per month and a tester is ₹ 60,000 per month. Average productivity for the team is 12 FP per person month. Which of the following represents the projected cost of the project?

- (A) ₹ 33,20,000 (B) ₹ 43,20,000
(C) ₹ 33,10,000 (D) ₹ 22,10,000

Q23 A complete n -ary tree is a tree in which each node has n children or no children. Let I be the number of internal nodes and L be the number of leaves in a complete n -ary tree. If $L = 41$, and $I = 10$, What is the value of n ?

- (A) 3 (B) 4
(C) 5 (D) 6

Q24 In a binary max heap containing n numbers, the smallest element can be found in ____ time.

- (A) $O(n)$ (B) $O(\log_2 n)$
(C) $O(1)$ (D) $O(\log_2 \log_2 n)$

Q25 If algorithm A and another algorithm B take $\log_2(n)$ and $\sqrt{n}$ microseconds, respectively, to solve a problem, then the largest size n of a problem these algorithms can solve, respectively, in one second are ____ and ____

- (A) 2^{10^6} and 10^6
(B) 2^{10^6} and 10^{12}
(C) 2^{10^6} and $6 \cdot 10^6$
(D) 2^{10^6} and $6 \cdot 10^{12}$

Q26 Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j if and only if either $j = i + 1$ or $j = 3i$. The minimum



number of edges in a path in G from vertex 1 to vertex 100 is ____

- (A) 23 (B) 99
(C) 4 (D) 7

Q27 Consider $L = L_1 \cap L_2$

Where

$$L_1 = \{0^m 1^m 20^n 1^n \mid m, n \geq 0\}$$

$$L_2 = \{0^m 1^n 2^k \mid m, n, k \geq 0\}$$

Then, the language L is

- (A) Recursively enumerable but not context free
(B) Regular
(C) Context free but not regular
(D) Not recursive

Q28 Let L_1 and L_2 be languages over $\Sigma = \{a, b\}$ represented by the regular expressions

$(a^* + b)^*$ and $(a + b)^*$ respectively.

Which of the following is true with respect to the two languages?

- (A) $L_1 \subset L_2$ (B) $L_2 \subset L_1$
(C) $L_1 = L_2$ (D) $L_1 \cap L_2 = \phi$

Q29 Which of the following statements is true?

- (A) The union of two context free languages is context free.
(B) The intersection of two context free languages is context free.
(C) The complement of a context free language is context free.
(D) If a language is context free, it can always be accepted by a deterministic pushdown automaton.

Q30 Which of the following is not an intermediate code form?

- (A) Syntax trees
(B) Three address codes
(C) Quadruples
(D) Post fix Notation.

Q31 Consider a code with only four valid code words: 0000000000, 0000011111, 1111100000, and 1111111111. This code has distance 5. If the code

word arrived is 0000000111 then the original code word must be ____

- (A) 0000011111 (B) 0000000000
(C) 1111100000 (D) 1111111111

Q32 Using 'RSA' public key cryptosystem, if

$P = 3$, $q = 11$ and $d = 7$, find the value of e and encrypt the number '19'.

- (A) 20,19 (B) 33,11
(C) 3,28 (D) 77,28

Q33 Protocols in which the sender sends one frame and then waits for an acknowledgement before proceeding for next frame are called as ____

- (A) Simplex protocols
(B) Unrestricted simplex protocols
(C) Simplex stop and wait protocols
(D) Restricted simplex protocols.

Q34 The period of a signal is 100 ms. Then the frequency of this signal in kilohertz is ____

- (A) 10 (B) 10^{-1}
(C) 10^{-2} (D) 10^{-3}

Q35 The process of removing details from a given state representation is called ____

- (A) Extraction (B) Mining
(C) Selection (D) Abstraction

Q36 Which of the following is NOT true in problem solving in artificial intelligence?

- (A) Implements heuristic search techniques
(B) Solution steps are not explicit
(C) Knowledge is imprecise
(D) It works on or implements repetition mechanism

Q37 If $f(x)=x$ is my friend, and $p(x)=x$ is perfect, then the correct logical translation of the statement "some of my friends are not perfect" is:

- (A) $\forall_x (f(x) \wedge \neg p(x))$
(B) $\exists_x (f(x) \wedge \neg p(x))$
(C) $\neg(f(x) \wedge \neg p(x))$
(D) $\exists_x (\neg f(x) \wedge \neg p(x))$

Q38 What kind of clauses are available in conjunctive normal form?



- (A) Disjunction of literals
 (B) Disjunction of variables
 (C) Conjunction of literals
 (D) Conjunction of variables.

Q39 Consider the following properties:

- A. Reflexive
 B. Antisymmetric
 C. Symmetric

Let $A = \{a, b, c, d, e, f, g\}$ and

$R = \{(a, a), (b, b), (c, d), (c, g), (d, g), (e, e), (f, f), (g, g)\}$

be a relation on (A). Which of the following property (properties) is (are) satisfied by the relation (R) ?

- (A) Only A (B) Only C
 (C) Both A and B (D) B and not A

Q40 Consider the following argument with premise $\forall x(P(x) \vee Q(x))$ and conclusion: $(\forall xP(x)) \wedge (\forall xQ(x))$

(A) $\forall x(P(x) \vee Q(x))$	Premise
(B) $P(c) \vee Q(c)$	Universal instantiation from (A)
(C) $P(c)$	Simplification from (B)
(D) $\forall xP(x)$	Universal generalization of (C)
(E) $Q(c)$	Simplification from (B)
(F) $\forall xQ(x)$	Universal generalization of (E)
(G) $(\forall xP(x)) \wedge (\forall xQ(x))$	Conjunction of (D) and (F)

- (A) This is a valid argument.
 (B) Steps (C) and (E) are not correct inferences
 (C) Steps (D) and (F) are not correct inferences
 (D) Step (G) is not a correct inference

Q41 Which of the following are applications of symbol table?

- (A) Storage allocation
 (B) Checking type compatibility
 (C) Suppressing duplicate error messages

Choose the correct answer from the options given below:

- (A) (A) and (B) only
 (B) (A) and (C) only

- (C) (B) and (C) only
 (D) (A), (B) and (C)

Q42 Simplified expression/s for following Boolean function

$$F(A, B, C, D) = \sum(0, 1, 2, 3, 6, 12, 13, 14, 15)$$

is/are

- (A) $A'B' + AB + A'C'D'$
 (B) $A'B' + AB + A'CD'$
 (C) $A'B' + AB + BC'D'$
 (D) $A'B' + AB + BCD'$

Choose the correct answer from the options given below:

- (A) (A) only
 (B) (B) only
 (C) (A) and (B) only
 (D) (B) and (D) only

Q43 Which of the following statements with respect to K-segment pipelining are true?

- (A) Maximum speedup that a pipeline can provide is k theoretically.
 (B) It is impossible to achieve maximum speedup k in k-segment pipeline.
 (C) All segments in pipeline take same time in computation.

Choose the correct answer from the options given below:

- (A) (A) and (B) only
 (B) (B) and (C) only
 (C) (A) and (C) only
 (D) (A), (B) and (C)

Q44 Which of the following statements with respect to multiprocessor system are true?

- (A) Multiprocessor system is controlled by one operating system.
 (B) In Multiprocessor system, multiple computers are connected by the means of communication lines.
 (C) Multiprocessor system is classified as multiple instruction stream and multiple data stream system.

Choose the correct answer from the options given below:



- (A) (A) only
- (B) (A) and (B) only
- (C) (A) and (C) only
- (D) (B) and (C) only

Q45 Which of the following statements regarding XML is/are True?

- (A) XML is a set of tags designed to tell browsers how to display text and images in a web page.
- (B) XML defines a syntax for representing data, but the meaning of data varies from application to application.
- (C) <Letter>, <LETTER>, and <letter> are three different tags in XML.

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A) and (C) only
- (C) (B) and (C) only
- (D) (A), (B) and (C)

Q46 In the context of 3D Computer graphics, which of the following statements is/are correct?

- (A) Under perspective projection, each set of parallel lines in the object do not stay parallel in the image (except those that are parallel to the viewplane to start with).
- (B) Applying a perspective transformation in the graphics pipeline to a vertex involves dividing by its 'z' coordinate.
- (C) Perspective transformation is a linear transformation.

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A) and (C) only
- (C) (B) and (C) only
- (D) (A), (B) and (C)

Q47 Concerning phong shading and Gouraud shading in a 3D scene, which of the following statements are true?

- (A) Gouraud shading requires more computation than phong shading.
- (B) Gouraud shading linearly interpolates the color of an interior pixel from the color at the

vertices.

(C) Phong shading interpolates over the normal vectors specified at the vertices.

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A) and (C) only
- (C) (B) and (C) only
- (D) (A), (B) and (C)

Q48 In the context of concurrency control, a given pair of operations in a schedule is called conflict schedule if

- (A) At least one of the operations is write operation.
- (B) Both the operations are performed on the same data item.
- (C) Both the operations are performed by different transactions.
- (D) Both the operations are performed on different data items.

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A), (B) and (C) only
- (C) (A), (C) and (D) only
- (D) (C) and (D) only

Q49 Software reliability is described with respect to

- (A) Execution Time
- (B) Calendar Time
- (C) Clock Time

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (B) and (C) only
- (C) (A), (B) and (C)
- (D) (A) and (C) only

Q50 To create an object-behavioral model, the analyst performs the following steps:

- (A) Evaluates all use-cases
- (B) Builds state transition diagram for the system.
- (C) Reviews the object behaviour model to verify accuracy and consistency



(D) Identifies events that do not derive the interaction sequence.

Choose the correct answer from the options given below:

- (A) (A), (B) and (C) only
- (B) (A), (B) and (D) only
- (C) (B), (C) and (D) only
- (D) (A), (C) and (D) only

Q51 Which of the following is/are behavioral testing technique(s)?

- (A) Equivalence Partitioning
- (B) Graph-Based Testing Method
- (C) Boundary Value Analysis
- (D) Data flow Testing
- (E) Loop Testing

Choose the correct answer from the options given below:

- (A) (B) and (D) only
- (B) (A), (B) and (C) only
- (C) (D) and (E) only
- (D) (A), (C) and (E) only

Q52 The running time of an algorithm is $O(g(n))$ if and only if:

- (A) its worst-case running time is $O(g(n))$ and its best-case running time is $\Omega(g(n))$.
- (B) its worst-case running time is $\Omega(g(n))$ and its best-case running time is $O(g(n))$.
- (C) $O(g(n)) = \Omega(g(n))$ ($O = \text{big } O$).
- (D) $o(g(n)) \cap \omega(g(n))$ is non-empty. ($o = \text{small } o$)

Choose the correct answer from the options given below:

- (A) (A) only
- (B) (B) only
- (C) (C) only
- (D) (D) only

Q53 Consider the following statements:

- (A) Any tree is 2-colorable
- (B) A graph G has no cycles of even length if it is bipartite.
- (C) A graph G is 2-colorable if it is bipartite.
- (D) A graph G can be colored with $d+1$ colors if d is the maximum degree of any vertex in the graph G .
- (E) A graph G can be colored with $O(\log|V|)$ colors if it has $O(|V|)$ edges.

Choose the correct answer from the options given below:

- (A) (C) and (E) are incorrect
- (B) (B) and (C) are incorrect
- (C) (B) and (E) are incorrect
- (D) (A) and (D) are incorrect

Q54 Which among the following statement(s) is (are) true?

- (A) A hash function takes a message of arbitrary length and generates a fixed length code.
- (B) A hash function takes a message of fixed length and generates a code of variable length.
- (C) A hash function may give same hash value for distinct messages.

Choose the correct answer from the options given below:

- (A) (A) only
- (B) (B) and (C) only
- (C) (A) and (C) only
- (D) (B) only

Q55 Let G_1 and G_2 be arbitrary context free languages and R an arbitrary regular language. Consider the following problems:

- (A) Is $L(G_1) = L(G_2)$?
- (B) Is $L(G_2) \leq L(G_1)$?
- (C) Is $L(G_1) = R$?

Which of the problems are undecidable?

Choose the correct answer from the options given below:

- (A) (A) only
- (B) (B) only
- (C) (A) and (B) only
- (D) (A), (B) and (C)

Q56 Consider the following languages:

$$L_1 = \{a^{zz} \mid z \text{ is an integer}\}$$

$$L_2 = \{a^{zz} \mid z \geq 0\}$$

$$L_3 = \{\omega\omega \mid \omega \in \{a, b\}^*\}$$

Which of the languages is (are) regular?

Choose the correct answer from the options given below:



- (A) L_1 and L_2 only
- (B) L_1 and L_3 only
- (C) L_1 only
- (D) L_2 only

Q57 Which of the following grammars is (are) ambiguous?

- (A) $s \rightarrow ss|asb|bsa | \lambda$
- (B) $s \rightarrow asbs | bsas | \lambda$
- (C) $s \rightarrow aAB$
 $A \rightarrow bBb$

$B \rightarrow A | \lambda$ where λ denotes empty string
Choose the correct answer from the options given below:

- (A) (A) and (C) only
- (B) (B) only
- (C) (B) and (C) only
- (D) (A) and (B) only

Q58 Which of the following cloud concept/s is/are related to pooling and sharing of resources?

- (A) Virtual Memory
- (B) Service
- (C) Virtualization

Choose the correct answer from the options given below:

- (A) (C) only
- (B) (A) and (B) only
- (C) (A) only
- (D) (B) only

Q59 Post office protocol (POP) is a message access protocol which is used to extract message for client. In this regard, which of the following are true?

- (A) POP has two modes, Delete mode and keep
- (B) In Delete mode, mail is deleted from mail box after each retrieval
- (C) In Delete mode, mail is deleted from mail box before each retrieval
- (D) In keep mode, mail is deleted before retrieval
- (E) In keep mode, mail remains in mail box after retrieval

Choose the correct answer from the options given below:

- (A) (A) and (B) only

- (B) (A), (D) and (E) only
- (C) (A), (B), (C) and
- (D) (A), (B) and (E) only

Q60 Firewall is a device that filters access to the protected network from the outside network. Firewalls can filter the packets on the basis of

- (A) Source IP Address
- (B) Destination IP Address
- (C) TCP Source Port
- (D) UDP Source Port
- (E) TCP Destination Port

Choose the correct answer from the options given below:

- (A) (A), (B) and (C) only
- (B) (B) and (E) only
- (C) (C) and (D) only
- (D) (A), (B), (C), (D) and (E) only

Q61 Consider the statement below.

A person who is radical (R) is electable (E) if he/she is conservative (C), but otherwise is not electable.

Few probable logical assertions of the above sentence are given below.

- (A) $(R \wedge E) \Leftrightarrow C$
- (B) $R \Rightarrow (E \Leftrightarrow C)$
- (C) $R \Rightarrow ((C \Rightarrow E) \vee \neg E)$
- (D) $(\neg R \vee \neg E \vee C) \wedge (\neg R \vee \neg C \vee E)$

Which of the above logical assertions are true?

Choose the correct answer from the options given below:

- (A) (B) only
- (B) (C) only
- (C) (A) and (C) only
- (D) (B) and (D) only

Q62 Which of the following statements are true?

- (A) A sentence α entails another sentence β if β is true in few worlds where α is true.
- (B) Forward chaining and backward chaining are very natural reasoning algorithms for knowledge bases in Horn form.
- (C) Sound inference algorithms derive all sentences that are entailed.



(D) Propositional logic does not scale to environments of unbounded size.

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (B) and (C) only
- (C) (C) and (D) only
- (D) (B) and (D) only

Q63 Which of the following statements are true?

- (A) Minimax search is breadth-first; it processes all the nodes at a level before moving to a node in next level.
- (B) The effectiveness of the alpha-beta pruning is highly dependent on the order in which the states are examined.
- (C) The alpha-beta search algorithm computes the same optimal moves as minimax algorithm.
- (D) Optimal play in games of imperfect information does not require reasoning about the current and future belief states of each player.

Choose the correct answer from the options given below:

- (A) (A) and (C) only
- (B) (A) and (D) only
- (C) (B) and (C) only
- (D) (C) and (D) only

Q64 Match List I with List II

Let $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$

List I	List II
(A) $R_1 \cup R_2$	$\{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$
(B) $R_1 - R_2$	$\{(1,1)\}$
(C) $R_1 \cap R_2$	$\{(1,2), (1,3), (1,4)\}$
(D) $R_2 - R_1$	$\{(2,2), (3,3)\}$

Choose the correct answer from the options given below:

- (A) A-I, B-II, C-IV, D-III
- (B) A-I, B-IV, C-III, D-II
- (C) A-I, B-III, C-II, D-IV
- (D) A-I, B-IV, C-II, D-III

Q65 Match List I with List II

List I	List II
(A) Handshaking	(I) I/O interface informs the CPU that device is ready for transfer
(B) Programmed I/O	(II) requires two control signals working in opposite directions
(C) Interrupt-initiated I/O	(III) has local memory & control large set of I/O devices.
(D) I/O processor	(IV) require CPU to check the I/O flag & perform transfer

Choose the correct answer from the options given below:

- (A) A-I, B-II, C-III, D-IV
- (B) A-II, B-IV, C-III, D-I
- (C) A-II, B-IV, C-I, D-III
- (D) A-IV, B-III, C-II, D-I

Q66 Match List-I with List-II. List I gives 3×3 matrices representing 2D transformations and List-II shows the corresponding transformation diagrams.

List I (Matrix)

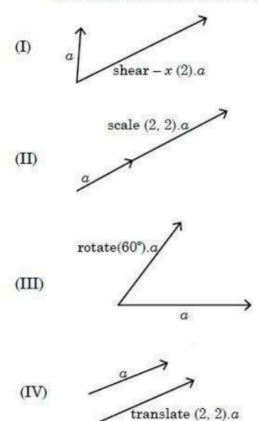
(A) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

List II (Transformation Diag)



Choose the correct answer from the options given below:

- (A) A-IV, B-II, C-III, D-I
- (B) A-IV, B-III, C-II, D-I
- (C) A-III, B-II, C-IV, D-I
- (D) A-II, B-IV, C-III, D-I

Q67



Match List I with List II.

List I (E-R symbols)	List II (Description)
(A)	(I) Key Attribute Type
(B)	(II) Weak Entity Type
(C)	(III) Total Participation of Entity in a relation
(D)	(IV) Multivalued Attribute type

Choose the correct answer from the options given below:

- (A) A-II, B-IV, C-III, D-I
 (B) A-IV, B-I, C-II, D-III
 (C) A-II, B-I, C-IV, D-III
 (D) A-III, B-IV, C-I, D-II

Q68 Match List I with List II

With reference to CMM developed by Software Engineering Institute (SEI)

List I	List II
(A) INITIAL	(I) Process measurement
(B) REPEATABLE	(II) Process definition
(C) DEFINED	(III) Project management
(D) MANAGED	(IV) ADHOC

Choose the correct answer from the options given below:

- (A) A-III, B-IV, C-II, D-I
 (B) A-IV, B-III, C-I, D-II
 (C) A-IV, B-III, C-II, D-I
 (D) A-III, B-IV, C-I, D-II

Q69 Match List I with List II

List I	List II
(A) Branch-and-bound	(I) Keeps track of all partial paths which can be candidate for further exploration.
(B) Steepest-descent hill climbing	(II) Detects difference between current state and goal state.
(C) Constraint satisfaction	(III) Discover problem state (s) that satisfy a set of constraints.

(D) Means-end-analysis	(IV) Consider all moves from current state and select best move.
------------------------	--

Choose the correct answer from the options given below:

- (A) A-I, B-IV, C-III, D-II
 (B) A-I, B-II, C-III, D-IV
 (C) A-II, B-I, C-III, D-IV
 (D) A-II, B-IV, C-III, D-I

Q70 Match List I with List II

List I	List II
(A) Topological sort of DAG	(I) $O(V+E)$
(B) Kruskal's MST algorithm	(II) $O(VE)$
(C) Bellman-Ford's single-source shortest path algorithm	(III) $\theta(V + E)$
(D) Floyd-Warshall's all pair shortest path algorithm	(IV) $\theta(V^3)$

Choose the correct answer from the options given below:

- (A) A-I, B-III, C-IV, D-II
 (B) A-III, B-I, C-IV, D-II
 (C) A-III, B-I, C-II, D-IV
 (D) A-I, B-III, C-II, D-IV

Q71 Match List I with List II:

L_R : Regular language, LCF: Context free language

L_{REC} : Recursive language, L_{RE} : Recursively enumerable language.

List I	List II
(A) Recursively Enumerable language	(I) $\overline{L_{REC}} \cup L_{RE}$
(B) Recursive language	(II) $\overline{L_{CF}} \cup L_{REC}$
(C) Context Free language	(III) $L_R^* \cap L_{CF}$

Choose the correct answer from the options given below:

- (A) A-II, B-III, C-I
 (B) A-III, B-I, C-II



(C) A-I, B-II, C-III

(D) A-II, B-I, C-III

Q72 Match List I with List II

List I	List II
(A) Serial Line IP (SLIP)	(I) Application layer
(B) Border Gateway Protocol (BGP)	(II) Transport layer
(C) User Data Protocol (UDP)	(III) Data Link layer
(D) Simple Network Management Protocol	(IV) Network layer

Choose the correct answer from the options given below:

(A) A-I, B-II, C-III, D-IV

(B) A-III, B-IV, C-II, D-I

(C) A-II, B-III, C-IV, D-I

(D) A-III, B-I, C-IV, D-II

Q73 Match List I with List II

List I	List II
(A) Greedy Best-First Search	(I) Space complexity is $O(d)$ where d =depth of the deepest optimal solution.
(B) A^*	(II) Incomplete even if the search space is finite.
(C) Recursive Best-First Search	(III) Optimal if the optimal solution is reachable; otherwise, returns the best reachable optimal solution.
(D) SMA^*	(IV) Computation and space complexity are too high.

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-I, D-III

(B) A-II, B-III, C-I, D-IV

(C) A-III, B-II, C-IV, D-I

(D) A-III, B-IV, C-II, D-I

Q74

Find the lexicographic ordering of the bit strings given below based on the ordering $0 < 1$.

(A) 001

(B) 010

(C) 011

(D) 0001

(E) 0101

Choose the correct answer from the options given below:

(A) $001 < 010 < 011 < 0001 < 0101$ (B) $0001 < 001 < 010 < 0101 < 011$ (C) $0001 < 0101 < 001 < 010 < 011$ (D) $001 < 010 < 0001 < 0101 < 011$

Q75 Arrange the following types of machine in descending order of complexity.

(A) SISD

(B) MIMD

(C) SIMD

Choose the correct answer from the options given below:

(A) A, B, C

(B) C, B, A

(C) B, C, A

(D) C, A, B

Q76 Given below are different properties of 3D projections from A-D. Identify the correct order on the basis of property true of (i) a perspective projection only, (ii) an orthographic projection only, (iii) both orthographic and projective transformations and (iv) neither orthographic nor projective transformation, respectively.

(A) Straight lines are mapped to straight lines.

(B) Distances and angles are (in general) preserved.

(C) Far away objects appear the same size as closer ones.

(D) Requires homogeneous coordinates in order for it to be encoded into a linear transformation.

Choose the correct answer from the options given below:

(A) D, C, B, A

(B) B, C, D, A

(C) D, C, A, B

(D) C, D, B, A

Q77 Arrange the following types of Cohesion from best to worst type.

(A) Logical Cohesion

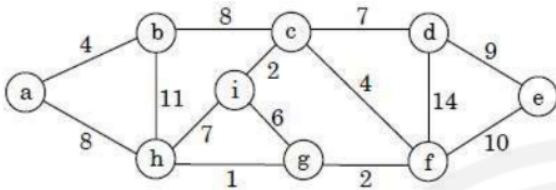


- (B) Sequential Cohesion
 (C) Communication Cohesion
 (D) Temporal Cohesion
 (E) Procedural Cohesion

Choose the correct answer from the options given below:

- (A) $A \rightarrow D \rightarrow E \rightarrow C \rightarrow B$
 (B) $A \rightarrow E \rightarrow D \rightarrow C \rightarrow B$
 (C) $B \rightarrow E \rightarrow C \rightarrow D \rightarrow A$
 (D) $B \rightarrow C \rightarrow E \rightarrow D \rightarrow A$

Q78 Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node a, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

- (A) (a,b), (a,h), (g,h), (f,g), (c,f), (c,i), (c,d), (d,e)
 (B) (a,b), (b,h), (g,h), (g,i), (c,i), (c,f), (c,d), (d,e)
 (C) (a,b), (b,c), (c,i), (c,f), (f,g), (g,h), (c,d), (d,e)
 (D) (a,b), (g,h), (g,f), (c,f), (c,i), (f,e), (b,c), (d,e)

Q79 Consider the following regular expressions:

- (a) $r = a(b + a)^*$
 (b) $s = a(a + b)^+$
 (c) $t = aa^*b$

Choose the correct answer from the options given below based on the relation between the languages generated by the regular expressions above:

- (A) $L(r) \subseteq L(s) \subseteq L(t)$
 (B) $L(r) \supseteq L(s) \supseteq L(t)$
 (C) $L(r) \supseteq L(t) \supseteq L(s)$
 (D) $L(s) \supseteq L(t) \supseteq L(r)$

Q80 Given below are two statements:

Statement I : 5 divides $n^5 - n$ whenever n is a nonnegative integer.

Statement II : 6 divides $n^3 - n$ whenever n is a nonnegative integer.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are correct
 (B) Both Statement I and Statement II are incorrect
 (C) Statement I is correct but Statement II is incorrect
 (D) Statement I is incorrect but Statement II is correct

Q81 Given below are two statements:

Statement I : Hardwired control unit can be optimized to produce fast mode of operation.

Statement II : Indirect addressing mode needs two memory reference to fetch the operand.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q82 Given below are two statements:

Statement I : Bezier curves are curves that interpolate all of their control points.

Statement II : A cubic bezier curve has four control points.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q83 Given below are two statements:

If two variables V_1 and V_2 are used for clustering, then consider the following

statements for k means clustering with $k = 3$

Statement I : If V_1 and V_2 have correlation of 1 the cluster centroid will be in straight line.

Statement II : If V_1 and V_2 have correlation of 0 the cluster centroid will be in straight line.

In the light of the above statements, choose the correct answer from the options given below



- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q84

<pre>/* P1 */ int main () { fork (); fork(); fork(); print("Happy\n"); }</pre>	<pre>/* P2 */ int main () { fork(); print("Happy\n"); fork(); print("Happy\n"); fork(); print("Happy\n "); }</pre>
--	--

Assuming that the system call `fork ()` never fails, consider the following C programs P1 and P2 executed on a UNIX / Linux system:

Statement I: P1 displays "Happy" 8 times.

Statement II: P2 displays "Happy" 12 times.

- In the light of the above statements, choose the correct answer from the options given below
- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q85 Given below are two statements:

Statement I: Quality control involves the series of inspections, reviews and tests used throughout the software process, to ensure each work product meets the requirements placed upon it.

Statement II : Quality assurance consists of auditing and reporting functions of management.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q86

Let G be a simple undirected graph, T_D be a DFS tree on G , and T_B be the BFS tree on G .

Consider the following statements.

Statement I : No edge of G is a cross with respect to T_D

Statement II : For every edge (u, v) of G , if u is at depth i and v is at depth j in T_B then $|i - j| = 1$.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q87 Given below are two statements:

Statement I : The problem "Is $L_1 \cap L_2 = \phi$?" is undecidable for context sensitive languages L_1 and L_2 .

Statement II : The problem "Is $W \in L$?" is decidable for context sensitive language L , (where W is a string).

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q88 Given below are two statements:

Statement I: The laws of nature put two fundamental limits on the data rate of a channel. The H.Nyquist limit restricts the number of independent samples per second to twice the bandwidth in a Noiseless channel.

Statement II: Shannon's major result about noised channel is that the maximum data rate of a channel whose bandwidth is H Hz, and whose signal-to-noise ratio is $\frac{S}{N}$ is given by:

Maximum number of bits/sec = $H \log_2(1 + \frac{S}{N})$

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true



- (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true.

Q89 Given below are two statements:

Statement I: In Ceasar Cipher each letter of Plain text is replaced by another letter for encryption.
 Statement II: The Diffie-Hellman algorithm is used for the exchange of secret keys.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true

Q90 Given below are two statements:

Statement I: A genetic algorithm is a stochastic hill-climbing search in which a large population of states is maintained.

Statement II: In nondeterministic environments, agents can apply AND-OR search to generate contingent plans that reach the goal regardless of which outcomes occur during execution.

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true

Directions (91-95) Read the following passage and answer the given questions.

Questions 91-95 concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.

- Q91** If **T** is the capacity of a track in bytes, and **S** is the capacity of each surface in bytes, then $(T, S) =$ _____
 (A) (50 K, 50000K)
 (B) (25 K, 25000K)

- (C) (25 K, 50000K)
 (D) (40 K, 36000K)

Q92 What is the capacity of the disk, in bytes?

- (A) 25,000 K (B) 500,000 K
 (C) 250,000 K (D) 50,000 K

Q93 Given below are two statements:

Statement I: The disk has a total number of 2000 cylinders.

Statement II: 51200 bytes is not a valid block size for the disk.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is correct but Statement II is false
 (D) Statement I is incorrect but Statement II is true

Q94 If the disk platters rotate at 5400 rpm (revolutions per minute), then approximately what is the maximum rotational delay?

- (A) 0.011 seconds (B) 0.11 seconds
 (C) 0.0011 seconds (D) 1.1 seconds

Q95 If one track of data can be transferred per revolution, then what is the data transfer rate?

- (A) 2,850 KBytes/second
 (B) 4,500 KBytes/second
 (C) 5,700 KBytes/second
 (D) 2,250 KBytes/second

Directions (96-100) Read the following passage and answer the given questions.

Consider the following table structures related to a university for the below Questions-

EMPLOYEE

NAME	VARCHAR (30)	NOT NULL
EID	VARCHAR (10)	NOT NULL



DEPTNO	INT (5)	NOT NULL
HODEID	VARCHAR (10)	
SALARY	INT (10)	

PRIMARY KEY (EID),

FOREIGN KEY (HODEID) REFERENCES EMPLOYEE (EID).

FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT (DID);

DEPARTMENT

DID	INT (5)	NOT NULL
DNAME	VARCHAR (30)	NOT NULL
HODID	VARCHAR (10)	NOT NULL
HODNAME	VARCHAR (30)	

PRIMARY KEY (DID),

UNIQUE (DNAME),

FOREIGN KEY (HODID) REFERENCES EMPLOYEE (EID);

PROJECT WORK

EMPID	VARCHAR (10)	NOT NULL
PROJNO	INT (5)	NOT NULL
PROJECTLOC	VARCHAR (30)	NOT NULL

PRIMARY KEY (EMPID, PROJNO),

FOREIGN KEY (EMPID) REFERENCES EMPLOYEE (EID),

Q96 On the basis of above given table structures, retrieve the distinct employee ID (EMPID) of all employees of university who are working on project No.20, 30 and 40.

- (A) SELECT EMPID
FROM PROJECTWORK
WHERE PROJNO=(20,30,40);
- (B) SELECT EMPID
FROM PROJECTWORK
WHERE PROJNO IN (20,30,40);
- (C) SELECT DISTINCT EMPID
FROM PROJECTWORK
WHERE PROJNO IN (20,30,40);
- (D) SELECT DISTINCT EMPID
FROM PROJECTWORK
WHERE PROJNO=20,30,40;

Q97

Given below are two statements to find the sum of salaries of all employees of the English department as well as the maximum, minimum and average salary in English department.

STATEMENT I : SELECT SUM (SALARY), MAX (SALARY), MIN (SALARY), AVG (SALARY) FROM EMPLOYEE, DEPARTMENT

WHERE DEPTNO=DID

AND DNAME='ENGLISH';

STATEMENT II: SELECT SUM (SALARY), MAX (SALARY), MIN (SALARY), AVG (SALARY) FROM EMPLOYEE, DEPARTMENT

WHERE DNAME='ENGLISH';

In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is correct but Statement II is false
- (D) Statement I is incorrect but Statement II is true

Q98 Which of the following query/queries return the employee ID and name of employees whose salary is greater than the salary of all employees in department number 20 of university? Order results by employee ID (refer to table structures given above).

- (A) SELECT EID, NAME
FROM EMPLOYEE
WHERE SALARY>(SELECT SALARY FROM EMPLOYEE WHERE DEPTNO=20)
ORDER BY EID;
- (B) SELECT EID, NAME
FROM EMPLOYEE WHERE SALARY>
(SELECT SALARY FROM EMPLOYEE WHERE DEPTNO =20);
- (C) SELECT EID, NAME



```
FROM EMPLOYEE
WHERE SALARY>ALL(SELECT SALARY
FROM EMPLOYEE WHERE DEPTNO =20)
ORDER BY EID
```

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A) and (C) only
- (C) (B) only
- (D) (C) only

Q99 In reference to the above given table structures, which of the following query/queries will drop the 'SALARY' column from 'EMPLOYEE' table?

- (A) ALTER TABLE EMPLOYEE DROP SALARY CASCADE;
- (B) ALTER TABLE EMPLOYEE DROP SALARY RESTRICT;
- (C) ALTER EMPLOYEE DROP SALARY;

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A) and (C) only
- (C) (B) and (C) only
- (D) (A) only

Q100 Refer table, structures given above. The university decided to give all employees in the 'SCIENCE' department a 20% rise in salary. Which of the following query/queries will compute the above results?

- (A) UPDATE EMPLOYEE
SET SALARY = SALARY * 1.20
WHERE DEPT NO. IN (SELECT DID FROM
DEPARTMENT WHERE
DNAME='SCIENCE');
- (B) UPDATE TABLE EMPLOYEE
SET SALARY = SALARY * 1.20
WHERE DNAME='SCIENCE';
- (C) ALTER TABLE EMPLOYEE
SET SALARY=SALARY * 1.20
WHERE DEPTNO. IN (SELECT DNAME
FROM DEPARTMENT WHERE DNAME =
'SCIENCE')

Choose the correct answer from the options given below:

- (A) (A) and (B) only
- (B) (A) only
- (C) (B) and (C) only
- (D) (C) only



Answer Key

Q1 (C)
Q2 (D)
Q3 (D)
Q4 (D)
Q5 (A)
Q6 (D)
Q7 (B)
Q8 (D)
Q9 (D)
Q10 (B)
Q11 (C)
Q12 (A)
Q13 (C)
Q14 (D)
Q15 (D)
Q16 (C)
Q17 (B)
Q18 (B)
Q19 (B)
Q20 (C)
Q21 (C)
Q22 (B)
Q23 (C)
Q24 (A)
Q25 (B)
Q26 (D)
Q27 (C)
Q28 (C)
Q29 (A)
Q30 (C)
Q31 (A)

Q32 (C)
Q33 (C)
Q34 (C)
Q35 (D)
Q36 (D)
Q37 (B)
Q38 (A)
Q39 (D)
Q40 (B)
Q41 (D)
Q42 (D)
Q43 (A)
Q44 (C)
Q45 (C)
Q46 (A)
Q47 (C)
Q48 (B)
Q49 (C)
Q50 (A)
Q51 (B)
Q52 (A)
Q53 (C)
Q54 (C)
Q55 (D)
Q56 (D)
Q57 (A)
Q58 (A)
Q59 (D)
Q60 (D)
Q61 (D)
Q62 (D)



Q63 (C)
Q64 (D)
Q65 (C)
Q66 (D)
Q67 (C)
Q68 (C)
Q69 (A)
Q70 (C)
Q71 (C)
Q72 (B)
Q73 (A)
Q74 (B)
Q75 (C)
Q76 (C)
Q77 (D)
Q78 (C)
Q79 (B)
Q80 (A)
Q81 (A)

Q82 (D)
Q83 (C)
Q84 (C)
Q85 (A)
Q86 (C)
Q87 (A)
Q88 (A)
Q89 (A)
Q90 (A)
Q91 (C)
Q92 (B)
Q93 (A)
Q94 (A)
Q95 (D)
Q96 (C)
Q97 (C)
Q98 (D)
Q99 (A)
Q100 (B)



Hints & Solutions

Q1 Text Solution:

The correct answer is **option C**.

To find the number of positive integers not exceeding 100 that are either odd or the square of an integer, We can divide this into two cases:

1. Odd numbers: There are 50 odd numbers from 1 to 100 (1, 3, 5, ..., 99).
2. Squares of integers: There are 10 perfect squares from 1 to 100 (1, 4, 9, 16, 25, 36, 49, 64, 81, 100).

However, we need to subtract the numbers that are both odd and perfect squares, as they are counted twice. There are 5 such numbers: 1, 9, 25, 49, and 81.

Therefore, the total number of positive integers not exceeding 100 that are either odd or the square of an integer is:

$$50 \text{ (odd numbers)} + 10 \text{ (perfect squares)} - 5 \text{ (numbers counted twice)} = 55$$

Q2 Text Solution:

Here, there are six copies of the same book into four identical boxes (same box). We will enumerate all ways to pack the books. For each way to pack the books, we will list the number of books in the box with the largest number of books, followed by the number of books in each box containing at least one book, in order to decrease the number of books in a box.

6,0,0,0
5,1,0,0
4,2,0,0
4,1,1,0
3,3,0,0
3,2,1,0
3,1,1,1
2,2,2,0
2,2,1,1

Q3 Text Solution:

The correct answer is **option D**.

D: $((p \wedge q) \rightarrow r)$ and $((p \rightarrow r) \wedge (q \rightarrow r))$

These two propositions are not logically equivalent. The first proposition states that if both p and q are true, then r is true. The second proposition states that if p is true, then r is true, and if q is true, then r is true. The first proposition requires both p and q to be true for r to be true, while the second proposition allows r to be true if either p or q is true.

Q4 Text Solution:

The correct answer is **option D**.

To find the optimum value of the linear programming (LP) problem, we can use the graphical method or the simplex method. Here, we will use the graphical method.

The constraints of the LP problem are:

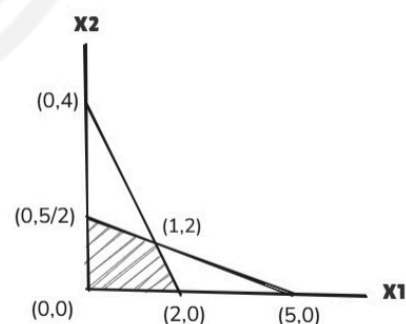
$$2x_1 + x_2 \leq 4, x_1 + 2x_2 \leq 5, (x_1, x_2 \geq 0)$$

We can graph these constraints on a coordinate plane, where x_1 is on the x-axis and x_2 is on the y-axis.

$$2x_1 + x_2 = 4 \text{ (constraint 1)}$$

$$x_1 + 2x_2 = 5 \text{ (constraint 2)}$$

$x_1 = 0$ (non-negativity constraint) and $x_2 = 0$ (non-negativity constraint)



The objective function is: $z = 2x_1 + 3x_2$

We want to maximize z , so we need to find the point in the feasible region that maximizes the value of z .

By inspecting the graph, we can see that the maximum value of z occurs at the point where the lines $2x_1 + x_2 = 4$ and $x_1 + 2x_2 = 5$ intersect. This point is $(1, 2)$.



Substituting $x_1 = 1$ and $x_2 = 2$ into the objective function, we get:

$$z = 2(1) + 3(2) = 2 + 6 = 8$$

Therefore, the optimum value of the LP is 8.

Q5 Text Solution:

The correct answer is **option A**.

Given:

Main memory size: 2^{16} bytes

Block size: 8 bytes

Cache size: 32 lines (direct mapped)

Since the cache is direct mapped, each line in the cache corresponds to a unique block in the main memory.

With 32 lines in the cache, we can address 2^5 (32) blocks in main memory.

Line: 5 bits (to address 32 lines in the cache)

The block size is 8 bytes, so we need 3 bits to address the 8 bytes within a block ($2^3 = 8$).

Word: 3 bits (to address 8 bytes within a block)

The remaining bits in the main memory address will be used for the Tag field, which identifies the block in the main memory.

Main memory size = 2^{16} bytes

Block size = 8 bytes = 2^3 bytes

Number of blocks in main memory = $2^{16} / 2^3 = 2^{13}$

Since we have 32 lines in the cache, we can address 2^5 blocks in main memory. To identify the remaining blocks, we need $2^{13} - 2^5 = 2^8$ blocks, which requires 8 bits.

Tag: 8 bits

Q6 Text Solution:

The correct answer is **option D**.

To determine the accumulator's content after the program's execution, we need to analyze the instructions and their effects on the accumulator. Here's a step-by-step execution of the program:

1. CLA (Clear Accumulator) at location 210:

Accumulator (A) is set to 0.

2. ADD 217 at location 211:

$$A = A + \text{contents of location 217} = 0 + 1234H = 1234H$$

3. INC (Increment) at location 212:

$$A = A + 1 = 1234H + 1 = 1235H$$

4. STA 217 (Store Accumulator) at location 213:

Contents of location 217 are updated to 1235H.

5. LDA 218 (Load Accumulator) at location 214:

A = contents of location 218 = 9CE2H

6. CMA (Complement Accumulator) at location 215:

$A = \sim A = \sim 9CE2H = 631DH$ (bitwise complement)

7. AND 217 at location 216:

$$A = A \wedge \text{contents of location 217} = 631DH \wedge 1235H = 0215H$$

After the execution of the program, the content of the accumulator is 0215H.

Q7 Text Solution:

The correct answer is **option B**.

Non-Pipeline System:

Time to process one task = 50 ns

Total time for 500 tasks = 500 tasks * 50 ns/task = 25000 ns

Pipelined System:

Clock cycle (T_p) = 10 ns

Number of segments in the pipeline (k) = 6

Total time for processing all tasks = $(n + k - 1) * T_p$

Total time = $(500 + 6 - 1) * 10 \text{ ns} = 5050 \text{ ns}$

Speedup Ratio:

$$\text{Speedup} = \frac{\text{Time taken by non-pipeline}}{\text{Time taken by pipeline}} = \frac{25000}{5050} \approx 4.95$$

Thus, the approximate speedup ratio of the pipeline for processing 500 tasks is 4.95.

Q8 Text Solution:

The correct answer is **option D**.

When solving the quadratic equation $ax^2 + bx + c = 0$, the sum of the roots can be calculated using the formula $-b/a$.

For the quadratic equation $x^2 - 10x + 26 = 0$ (where $a=1$ and $b=-10$), we know that the roots are 4 and 7. Therefore, the sum of the roots is given by:

$$\text{the sum of roots} = \frac{-b}{a} = \frac{(10)_r}{(1)_r} = 4_r + 7_r$$

$$r = 4 + 7 = 11$$

Q9 Text Solution:

The correct answer is **option D**.



Let's analyze the pseudo-code fragment:

1. Initialization: `x` starts at 0. `p` starts at 1.
 2. Loop Condition: The while loop runs while `p < b`.
 3. Loop Execution: Each iteration multiplies `p` by `a` and increments `x` by 1.
- Since `a > 1` and `a < b`, `p` will eventually exceed `b`, causing the loop to terminate.
- At termination, we have:

- Initially: `x = 0`, `p = 1`.
- After 1st iteration: `p = a`, `x = 1`.
- After 2nd iteration: `p = a^2`, `x = 2`.
- After the k-th iteration: `p = a^k`, `x = k`.

The loop ends when `p >= b`. Since `a > 1`, `a^k` grows exponentially, meaning there exists a smallest integer `k` such that `a^k >= b`.

Thus, the value of `x` at termination is: $x = \lceil \log_a(b) \rceil$

where $\lceil x \rceil$ represents the ceiling function, which rounds up to the nearest integer.

Q10 Text Solution:

The correct answer is **option B**.

B: defining the clickable region in an image

The <map> tag in HTML is used in conjunction with the <area> tag to define clickable areas (regions) within an image. This allows specific parts of an image to link to different destinations.

Q11 Text Solution:

To determine the total space consumed by the array A[10], we need to calculate the size of each struct element and then multiply it by 10.

Since the compiler is not permitted to reorder fields, we need to add padding to ensure alignment. The alignment rules require the address of every primitive data element to be an integer multiple of the element's size.

Here's the padded struct:

short s: 2 bytes (aligned), it occupies 0 and 1 byte
char c: 1 byte (not aligned, needs padding), so it is placed at 2nd byte

Add 1 byte of padding to align short t to a 2-byte boundary,

short t: 2 bytes (aligned), so it occupies 4 and

5th byte

char d: 1 byte (not aligned, needs padding), so it is placed at 6th byte

Add 1 byte of padding to align double r to an 8-byte boundary

double r: 8 bytes (aligned), It starts from 8 and spans to 15th byte

int i: 4 bytes (aligned), it is placed at 16 to 19th bytes and the bytes from 20 to 23 are wasted

The total size of each struct element is:

2 (s) + 1 (c) + 1 (padding) + 2 (t) + 1 (d) + 1 (padding) + 8 (r) + 4 (i) + 4 = 24 bytes

Since we have an array of 10 structs, the total space consumed is:

24 bytes/struct × 10 structs = 240 bytes

Therefore, the array A[10] will consume 240 bytes of memory.

Q12 Text Solution:

Option A is Correct

1. First Recursion Call: f(1,3)

- result=1/3=0.3333
- Recursive call: f(2,9)

2. Second Recursion Call: f(2,9)

- result=2/9=0.2222
- Recursive call: f(4,27)

3. Third Recursion Call: f(4,27)

- Result = 4/27 = 0.1481
- Recursive call: f(8,81)

4. Fourth Recursion Call: f(8,81)

- result=8/81=0.0988
- Recursive call: f(16,243)

5. Fifth Recursion Call: f(16,243)

- result=16/243=0.0658
- Recursive call: f(32,729)

6. Sixth Recursion Call: f(32,729)

- result=32/729=0.0439
- Recursive call: f(64,2187)

7. Seventh Recursion Call: f(64,2187)

- result=64/2187=0.0293



- Recursive call: $f(128,6561)$

8. Eighth Recursion Call: $f(128,6561)$

- result = $1286561 = 0.0195$
- Recursive calls continue, but contributions become negligible.

Summing the values and applying the recursive subtraction:

$f(1,3) =$

$0.3333 - 0.2222 + 0.1481 - 0.0988 + 0.0658 - 0.0439 + 0.0293 - \dots$

Approximating this series: $f(1,3) \approx 0.2$

Q13 Text Solution:

The correct answer is **option C**.

C: Worker Node and Master Node, respectively

In Hadoop, the NameNode is the master node that manages the filesystem namespace and controls client access to files, while the DataNode is the worker node that stores the actual data. Therefore, the NameNode is considered the master node, and the DataNode is considered the worker node.

Name Node = Master Node, Data Node = Worker Node

Q14 Text Solution:

The correct answer is **option D**.

A candidate key is a set of one or more attributes in a database table that can uniquely identify each record in the table.

$(UYZ)^+ = (U,V,W,X,Y,Z)$

$(VYZ)^+ = (U,V,W,X,Y,Z)$

$(XYZ)^+ = (U,V,W,X,Y,Z)$

Q15 Text Solution:

To determine the number of page faults incurred by the FIFO (First-In-First-Out) and LRU (Least Recently Used) page replacement algorithms, we will simulate both algorithms using the provided page reference string: $\langle A, B, C, D, A, B, E, A, B, C, D, E, B, A, B \rangle$.

1) **FIFO:**

A	B	C	D	A	B	E	A	B	C	D	E	B	A	B
A	A	A	D	D	D	E			E	E		B	B	
	B	B	B	A	A	A			C	C		C	A	
		C	C	C	B	B			B	D		D	D	

Page fault = 11

2) **LRU:**

A	B	C	D	A	B	E	A	B	C	D	E	B	A	B
A	A	A	D	D	D	E			C	C	C	B	B	
	B	B	B	A	A	A			A	D	D	D	A	
		C	C	C	B	B			B	B	E	E	E	

Page fault = 12

Q16 Text Solution:

The correct answer is **option C**.

To determine the time taken to satisfy all the disk requests using the Shortest Seek Time First (SSTF) algorithm, we need to process the requests in an order that minimizes the total seek time.

Given:

Current position of the disk head: 20

Disk requests: 10, 22, 20, 2, 40, 6, 38

Time to move from one cylinder to the adjacent one: 2 milliseconds

Execution:

Current position: 20

- Closest request: 20 (0 distance)
Move to 20: 0 milliseconds (already there)
Current position: 20
- Remaining requests: 10, 22, 2, 40, 6, 38
Closest request: 22 (2 distance)
Move to 22: (2 ms times 2 = 4 ms)
Current position: 22
- Remaining requests: 10, 2, 40, 6, 38
Closest request: 20 (already served)
Closest request: 10 (12 distance)
Move to 10: (2 ms times 12 = 24 ms)
Current position: 10
- Remaining requests: 2, 40, 6, 38
Closest request: 6 (4 distance)
Move to 6: (2 ms times 4 = 8 ms)
Current position: 6
- Remaining requests: 2, 40, 38
Closest request: 2 (4 distance)



Move to 2: (2 ms times 4 = 8 ms)

Current position: 2

- Remaining requests: 40, 38
Closest request: 38 (36 distance)
Move to 38: (2 ms times 36 = 72 ms)
Current position: 38
- Remaining request: 40
Move to 40: (2 ms times 2 = 4 ms)

Total Time = 0 + 4 + 24 + 8 + 8 + 72 + 4 = 120 ms

Q17 Text Solution:

Option B is Correct

To calculate the largest file that can be represented by an i-node in the given Linux file system. The given Key Parameters are:

1. Block Size: 2KB = 2048bytes
2. Disk Address Size: 32bits = 4bytes
3. i-node structure:
 - Contains 12 direct block addresses.
 - 1 single indirect block (contains addresses pointing to blocks).
 - 1 double indirect block (contains addresses pointing to other blocks that contain block addresses).

Each direct block stores 2KB, and there are 12 direct blocks:

Storage from direct blocks = $12 \times 2\text{KB} = 24\text{KB}$.

The single indirect block contains pointers to blocks. Each pointer is 4 bytes, so the number of pointers that fit in a block is:

Pointers per block = $\text{Block Size} / \text{Pointer Size} = 2048 / 4 = 512$ pointers.

Each pointer points to a 2KB block, so the total storage from the single indirect block is:

Storage from single indirect block = $512 \times 2\text{KB} = 1024\text{KB} = 1\text{MB}$.

The double indirect block contains pointers to single indirect blocks. Each single indirect block can address 512 data blocks. Thus, the number of data blocks addressable by the double indirect block is:

Data blocks from double indirect block = $512 \times 512 = 262144$ blocks

Each block is 2KB, so the total storage from the double indirect block is:

Storage from double indirect block = $262144 \times 2\text{KB} = 524288\text{KB} = 512\text{MB}$

Adding the contributions from direct blocks, single indirect block, and double indirect block:

Total storage = $24\text{KB} + 1\text{MB} + 512\text{MB} = 513\text{MB}$

Q18 Text Solution:

Option B is Correct

To calculate the Effective Memory Access Time (EMAT) in a system with a single-level page table and a TLB (Translation Lookaside Buffer), we use the formula:

$\text{EMAT} = (\text{Hit Rate} \times \text{Hit Time}) + (\text{Miss Rate} \times \text{Miss Time})$

Given Data:

1. Hit rate to TLB: 80% (0.8)
2. TLB search time: 15 ns
3. Memory access time: 150 ns
4. Miss rate: $1 - \text{Hit Rate} = 0.2$

Memory access details:

On a TLB hit: Memory is accessed once (150 ns), TLB search adds 15 ns.

So, Total time = $15 + 150 = 165\text{ns}$

On a TLB miss: Memory is accessed twice:

- First, to fetch the page table entry (150 ns).
- Second, to fetch the data (150 ns).

TLB search adds 15 ns. So, Total time = $15 + 150 + 150 = 315\text{ns}$

Calculating EMAT, $\text{EMAT} = (\text{Hit Rate} \times \text{Hit Time}) + (\text{Miss Rate} \times \text{Miss Time})$

$\text{EMAT} = (0.8 \times 165) + (0.2 \times 315) = 132 + 63 = 195\text{ns}$.

Q19 Text Solution:

The correct answer is **option B**.

Use-Case diagrams provide a static view of a system by showing the interactions between users (actors) and the system, representing the functional requirements. They depict the system's functionalities without detailing the dynamic behavior or the sequence of operations.

Q20 Text Solution:

The correct answer is **option C**.



Perfective maintenance : Perfective maintenance involves modifying the software to improve performance or enhance features without fixing defects. This type of maintenance often includes restructuring the code to make it more efficient or easier to understand.

Q21 Text Solution:

Option- **C(3000)**

Explanation:

Key Information: - LOC in L2 = 3 x LOC in L1
- We need to identify the LOC in L1 where the overall cost (development + maintenance) for both languages is equal. We need to determine the moment at which the overall cost of using L1 equals the total cost of utilizing L2.

Total Cost = Development Cost + Maintenance Cost

Total cost for L1: $L1 = (L1/1000) * (70,000) + (10 * 1,00,000) = 70 L1 + 1,00,0000$

Total Cost for L2: $L2 = (L2/1000) * (90,000) + (10 * 40,000) = 90 L2 + 4,00,000$

Given, **L2 = 3L1**

$$90 L2 + 4,00,000 = 70 L1 + 1,00,0000$$

$$270 L1 + 4,00,000 = 70 L1 + 1,00,0000$$

$$L1 = 6,00,000 / 200 = 3000$$

Q22 Text Solution:

Option- **B(4,320,000)**

Explanation:

1. Total Effort:

$$\frac{864 FP}{12 FP / Person-month} = 72 person - months.$$

$$\mathbf{2. Duration:} \quad \frac{72 person-month}{6 people} = 12 months$$

3. Total Monthly Cost:

$$\begin{aligned} & ₹50,000(\text{Requirement Gatherer}) + \\ & ₹70,000(\text{Designer}) + ₹120,000(2 \text{ Programmers}) + \\ & ₹120,000(2 \text{ Testers}) = ₹360,000. \end{aligned}$$

$$\mathbf{4. Total Cost:} \quad ₹360,000 \times 12 \text{ months} = ₹4,320,000.$$

Q23 Text Solution:

Option- **C(5)**

Explanation: In a complete n-ary tree, the relationship between the number of internal nodes I and leaves L is provided by: $L = I \cdot (n-1) + 1$

Step 1: Substitute the specified values: We have been given: $L = 41, I = 10$

Substituting these values into the equation: $41 = 10 (n - 1) + 1$

Step 2: Simplify the equation: $41 - 1 = 10 (n - 1)$

$$40 = 10 (n - 1)$$

$$n - 1 = 40/10$$

$$n - 1 = 4$$

$$n = 4 + 1 = 5$$

Q24 Text Solution:

Option- **A(O(n))**

Explanation: In a binary max heap, the smallest element is found at one of the leaf nodes, as all parent nodes have values greater than or equal to their children. This process requires traversing all leaf nodes, requiring $O(n)$ time in the worst-case scenario.

The smallest element in a binary max heap is found among the leaf nodes, which is approximately $[n/2]$. Scanning them takes $O(n)$ time.

Q25 Text Solution:

Option- **B(2^{10^6} and 10^{12})**

Explanation:

Algorithm A: Given, $(\log)_2 n \leq 10^6$

Problem solution time is $(\log)_2 n$ microseconds. So in 1 second, the maximum value of n such that $\log_2 n \leq 10^6$: Converting to 10 base-log: $n \leq 2^{10^6}$

Algorithm B: Given, $\sqrt{n} \leq 10^6$

Problem-solving time is $\sqrt{n}$ microseconds or

$(1000000)^2$. So, in one second, the maximum value of n: Squaring both sides yields: $n \leq (1000000)^2$, $n = 10^{12}$

Q26 Text Solution:

Option- **D(7)**

Explanation: We aim to reach 100 from 1, with edges defined as:

$j=i+1$ (increment by 1) or $j=3i$ (multiply by 3).



Verifying the Path: Since we have to find minimum values, so we need to make an edge that has maximum difference in i and j . Depending on it we will take the first or second choice.

Let us follow a multiple of three. After $1 \Rightarrow 3 \Rightarrow 9 \Rightarrow 27 \Rightarrow 81$, we cannot continue with multiples of 3.

So we'll have to use $i+1$. This solution leads to a long path.

If the value is not a multiple of three, consider starting from the end and reducing by one.

Otherwise, divide by three.

$100 \Rightarrow 99 \Rightarrow 33 \Rightarrow 11 \Rightarrow 10 \Rightarrow 9 \Rightarrow 3 \Rightarrow 1$

$100-1=99$

$99-1=98$, not a multiple of 3. Thus, divide 99 by 3= 33

$33-1=32$, not a multiple of 3. Thus, divide 33 by 3= 11

$11-1=10$, not a multiple of 3. Thus, divide 11 by 3= 3.6(not possible). So, we apply $j=j-1$ here.

$10-1=9$, Multiple of 3

$9-1=8$, not a multiple of 3. Thus, divide 9 by 3= 3

$3-1=2$, not a multiple of 3. Thus, divide 3 by 3= 1

Counting the Edges: Each step corresponds to one edge:

$1 \rightarrow 3, 3 \rightarrow 9, 9 \rightarrow 10, 10 \rightarrow 11, 11 \rightarrow 33, 33 \rightarrow 99, 99 \rightarrow 100$

Total edges: 7

Q27 Text Solution:

Option- **C(Context-free but not regular)**

Explanation: Given, the language $L = L_1 \cap L_2$ where:

$L_1 = \{0^m 1^n 2^0 0^n 1^n | m, n \geq 0\}$ which requires two blocks of $0^m 1^n$ (context-free).

$L_2 = \{0^m 1^n 2^k | m, n, k \geq 0\}$, which is regular.

The intersection L results in strings of form $0^m 1^{2m} 2 | m \geq 0$, which is context-free but not regular.

Q28 Text Solution:

Option- **C($L_1=L_2$)**

Explanation: $(a+b)^*$ is a universal set grammar capable of producing all strings using all conceivable alphabets "a" and "b".

$(a^* + b)^* = a^*$ can yield epsilon and the remaining half is similar to $(a + b)^*$

$(a^* + b)^* = (a + b^*)^* = (a^* b^*)^* = (a^* + b^*)^* = a^* (ba^*)^* = b^* (ab^*)^* = (a + b)^*$ all grammar are equal. Hence the Hence, $L_1 = L_2$.

Q29 Text Solution:

Option- **A(The union of two context-free languages is context-free)**

Explanation: Yes, combining two context-free languages (CFLs) is context-free. This is because the union closes context-free languages. If L_1 and L_2 are CFLs, a new context-free grammar can be created by introducing a new start symbol and adding rules to derive either L_1 or L_2 . Hence, $L_1 \cup L_2$ is also context-free.

Q30 Text Solution:

Option- **C(Quadruples)**

Explanation: Intermediate code can be represented in three ways: postfix notation, syntax trees, and three address codes.

There are three types of representations: quadruples, triples, and indirect triples.

Quadruples: These are a type of intermediate representation used in compilers, with each instruction having four fields: operator, two operands, and a result.

Q31 Text Solution:

Option- **A(0000011111)**

Explanation: Arrange the XOR procedures below each other.

To rectify this, the d-bit error hamming distance should be $2d+1$.

So, $2d+1=5$ $d=2$

The hamming distance between the code word 0000000111 and the given choice should be two bits. Only option 1 is applicable.

(xor) 00000001 11	(xor) 00000001 11	(xor) 00000001 11	(xor) 00000001 11
000001111 1	00000000 00	111110000 0	111111111 1111111111
00000110	00000011	1111110011	1111111000



00 (2-bit error)	10 (3-bit error)	(7-bit error)	(8-bit error)
------------------------	------------------------	------------------	------------------

0000011111 is minimum with 2-bit error.

Q32 Text Solution:

Option: **C(3,28)**

Explanation: Given: $p=3$, $q=11$, $d=7$. We need to find e and encrypt the number 19.

Steps:

1. Calculate n : $n=p \times q=3 \times 11=33$
2. Calculate $\phi(n)$ using Euler's Totient function:
 $\phi(n) = (p-1)(q-1) = 2 \times 10 = 20$
3. Find e such that: $d \times e \equiv 1 \pmod{20}$
 Given $d=7$, we get that $e=3$ (since $7 \times 3 = 21 \pmod{20} \equiv 1$).
4. Encrypt the number 19. To encrypt, we use this formula:

$$c = 19^3 \pmod{33}$$

$$19^3 = 6859, \text{ and } 6859 \pmod{33} = 28$$

The public key exponent(e) is 3 and the encrypted message is 28

Q33 Text Solution:

Option: **C(Simplex stop and wait protocols)**

Explanation:

In Simplex Stop-and-Wait protocols, senders send one frame at a time and wait for an acknowledgment (ACK) from the receiver before sending the next frame. This guarantees that each frame is validated before moving on to the next.

Q34 Text Solution:

Option- **C(10^{-2})**

Explanation: Given:- Time Period, $T = 100\text{ms}$
 We need to compute the frequency in kilohertz (kHz).

Step 1: Convert the period to seconds. $100\text{ ms} = 100/10^3\text{ seconds} = 0.1\text{ seconds}$

Step 2: Determine the periodicity:

$$f = 1/T = 1/0.1 = 10\text{ Hz}$$

Step 3: Convert frequency to Kilohertz $1\text{ Hz} = 10^{-3}\text{ kHz}$

$$f = 10\text{ Hz} = 10 \times 10^{-3}\text{ kHz} = 10^{-2}\text{ kHz}$$

Q35 Text Solution:

Abstraction is the process of hiding unnecessary details and focusing on the essential features of a system or concept. It simplifies complex systems by reducing them to their core components, making it easier to work with or understand.

Option D is correct answer.

Q36 Text Solution:

In problem-solving in artificial intelligence:

1. **Heuristic search techniques** are widely used to solve problems (e.g., BFS, DFS, A*, Best-First Search).
2. **Solution steps are not explicit** as the AI discovers them dynamically during the search.
3. **Knowledge is imprecise**, especially in real-world applications.

Not all AI problem-solving approaches require repetition mechanisms. While many algorithms (e.g., BFS, DFS) involve iterative processes, some direct heuristic methods may derive solutions without repeating steps.

The statement "It works on or implements repetition mechanism" is **not true**.

Q37 Text Solution:

$f(x)=x$ means "x is my friend."

$p(x)=x$ means "x is perfect."

Logical Translation:

"Some of my friends" implies the existence of at least one friend. This can be represented using the existential quantifier $\exists x$, which means "there exists at least one x."

The phrase "are not perfect" means that for this particular x, the predicate $p(x)$ (i.e., "x is perfect") does not hold. This can be expressed as $\neg p(x)$, meaning "x is not perfect."

The correct translation of "Some of my friends are not perfect" is:

$$\exists x(f(x) \wedge \neg p(x))$$

Option B is correct.

Q38 Text Solution:

Conjunctive Normal Form (CNF) is a way of structuring logical expressions in propositional



logic. A formula needs to be expressed as a conjunction (AND) of one or more disjunctions (OR) of literals.

- A literal is a variable or its negation (e.g., A , $\neg A$).
- A clause is a disjunction (OR) of literals (e.g., $AV-BVC$).
- In CNF, multiple clauses are combined using conjunction (AND): $(AV-B) \wedge (\neg CVD)$

A. Disjunction of literals:

Correct. Each clause in CNF is a disjunction (OR) of literals. For example, $AV-B$ is a valid clause in CNF.

Q39 Text Solution:

1. Reflexive:

A relation R is **reflexive** if $(x,x) \in R$ for all $x \in A$. In R , $(a,a), (b,b), (e,e), (f,f)$, and (g,g) are included. For c and d , (c,c) and (d,d) are **not** in R .

R is **not reflexive**.

2. Antisymmetric:

A relation R is **antisymmetric** if for all $x, y \in A$, whenever $(x,y) \in R$ and $(y,x) \in R$, it must be that $x=y$.

For R :

No pair (x,y) and (y,x) exists simultaneously (e.g., $(c,d) \in R$, but $(d,c) \notin R$).

R satisfies the **antisymmetric** property.

3. Symmetric:

A relation R is **symmetric** if for all $x, y \in A$, whenever $(x,y) \in R$, $(y,x) \in R$ must also hold.

For R :

$(c,d) \in R$, but $(d,c) \notin R$. So, $(c,g) \in R$, but $(g,c) \notin R$.

R is **not symmetric**.

The correct option is:

D. B and not A

Q40 Text Solution:

For **C** and **E** to be true the assertion should be $P(c) \wedge Q(c)$.

Q41 Text Solution:

(A) Storage allocation:

Valid: The symbol table keeps track of variable names and their attributes, such as data type, size, and memory location.

This information allocates memory for variables and functions during compilation.

(B) Checking type compatibility:

Valid: The symbol table stores the data types of variables and functions. During semantic analysis, the compiler checks whether operations involving variables are type-compatible.

For instance, the symbol table helps identify errors when adding a string to an integer.

(C) Suppressing duplicate error messages:

Valid: When the compiler encounters the same error multiple times (e.g., multiple uses of an undeclared variable), the symbol table prevents redundant error reporting.

The symbol table records declared symbols and avoids repeated errors for the same issue.

Option D is correct.

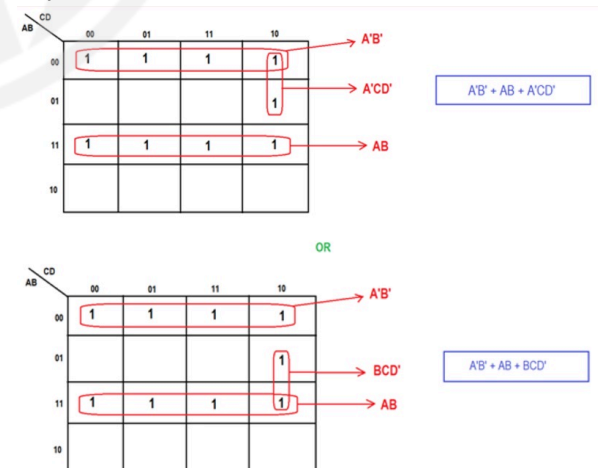
Q42 Text Solution:

$$F(A,B,C,D) = \sum(0,1,2,3,6,12,13,14,15)$$

The function is a sum (OR) of the Boolean expressions corresponding to these minterms.

Each number corresponds to a binary representation of A, B, C , and D . Each bit in the binary number represents the value of one of the variables A, B, C , and D (0 for false, 1 for true).

Using K-Map we can get the following expressions:



Q43 Text Solution:

(A) Maximum speedup that a pipeline can provide is k theoretically.

True.



In an ideal scenario, the speedup of a pipeline is equal to the number of segments k . This happens if the pipeline is fully loaded with no delays (i.e., no stalls, hazards, or latency issues).

(B) It is impossible to achieve maximum speedup k in k -segment pipeline.

True.

Due to pipeline hazards (structural, data, and control hazards) and other overheads (e.g., time required for pipeline latching and flushing), achieving the theoretical maximum speedup k is not possible.

Option A is correct.

Q44 Text Solution:

A. Multiprocessor system is controlled by one operating system.

- True: A multiprocessor system is a tightly coupled system where multiple processors share the same memory and are controlled by a single operating system. The OS manages processes, memory, and coordination between the processors.

C. Multiprocessor system is classified as multiple instruction stream and multiple data stream system.

- True: A multiprocessor system follows the **MIMD (Multiple Instruction stream, Multiple Data stream)** model. In this model, each processor executes different instructions on different data streams simultaneously, characteristic of multiprocessor systems.

Option C is the answer.

Q45 Text Solution:

(B) XML defines a syntax for representing data, but the meaning of data varies from application to application.

True.

XML is a markup language that provides a structure for data representation. However, the meaning of the tags and the data is application-specific and defined by the developer.

(C) <Letter>, <LETTER>, and <letter> are three different tags in XML.

True.

XML is case-sensitive, meaning tags with different cases are treated as entirely separate tags.

Option C is correct.

Q46 Text Solution:

Perspective Projection:

Simulates the way the human eye perceives depth, causing objects farther from the viewer to appear smaller.

A key property is that lines that are parallel in the 3D world may appear to converge in the image, forming a vanishing point, except for lines parallel to the view plane.

Perspective Transformation:

Transforms 3D points into a 2D perspective view. This transformation involves dividing the coordinates by the depth (**z-coordinate**) to achieve the perspective effect.

Linearity of Perspective Transformation:

Perspective transformation is **not a linear transformation** because dividing by the z -coordinate introduces a non-linear component. Linear transformations maintain straight-line relationships, while perspective projection does not.

Option A is correct.

Q47 Text Solution:

(B) Gouraud shading linearly interpolates the color of an interior pixel from the color at the vertices.

True. This is the core principle of Gouraud Shading, where the color at each interior pixel is calculated by linearly interpolating the colors of the vertices.

(C) Phong shading interpolates over the normal vectors specified at the vertices.

True. Phong Shading interpolates the surface normal vectors across the polygon and uses these interpolated normals to compute the color at each pixel.

Option C is correct.



Q48 Text Solution:

(A) At least one of the operations is a write operation:

This is a necessary condition for a conflict, as two-read operations do not cause conflicts.

(B) Both the operations are performed on the same data item:

This is essential because conflicts occur only when the same data item is accessed by different operations.

(C) Both the operations are performed by different transactions:

This is required because operations within the same transaction are always conflict-free.

Option B is correct.

Q49 Text Solution:

Software reliability refers to the probability of a software system operating without failure under specified conditions for a specified period. It is measured using various time-related factors that affect the system's operation.

Execution Time (A):

This means the actual time the software is in use or running.

Reliability is measured during execution because it directly reflects the operational performance of the software.

Calendar Time (B):

This represents the elapsed time from deployment or release, including downtime or periods when the software is not actively in use. While it can indicate reliability trends over a long term (e.g., maintenance issues or failure frequency), it is not a primary factor for real-time reliability.

Clock Time (C):

It refers to real-world time during which the software operates, typically including idle times or when the software is not executing actively. Reliability during clock time is also relevant, especially for systems that depend on uptime and availability.

Option C is correct.

Q50 Text Solution:

An object-behavioral model in software engineering defines how objects interact and behave in response to events within the system.

Steps:

Evaluate All Use-Cases (A):

Use cases help identify how objects interact based on user scenarios and system functionality. This is a step in creating an object-behavioral model.

Build State Transition Diagram for the System

(B):

A state transition diagram shows how an object changes state in response to events, capturing the behavioral dynamics.

This is a step for modeling object behavior.

Review the Object Behavior Model for Accuracy and Consistency (C):

Reviewing the model ensures it is consistent and accurate with the requirements and system design.

This step is part of refining the object-behavioral model.

Option A is correct answer.

Q51 Text Solution:

Behavioral testing focuses on the external behavior of the software application based on its specifications. It is a black-box testing technique.

Techniques:

Equivalence Partitioning (A):

Divides the input data into equivalent partitions to reduce the number of test cases.

It is a behavioral testing technique.

Graph-Based Testing Method (B):

Models the software behavior using graphs such as state diagrams or control flow graphs.

It is a behavioral testing technique.

Boundary Value Analysis (C):

Focuses on testing the boundaries of equivalence partitions.

It is a behavioral testing technique.

Data Flow Testing (D):

Tracks the flow of data through the code and identifies anomalies.

It is a **structural testing technique**



Loop Testing (E):

Examines loops in the control structure of the code.

It is a **structural testing technique**.

Option B is correct.

Q52 Text Solution:

Option A: "Its worst-case running time is $O(g(n))$ and its best-case running time is $\Omega(g(n))$ "

- Worst-case running time is $O(g(n))$, the algorithm cannot exceed $g(n)$ in time.
- Best-case running time is $\Omega(g(n))$, then even the best-case time is bounded below by $g(n)$, these conditions ensure the running time is in the expected $O(g(n))$ range.

Q53 Text Solution:

(B) A graph G has no cycles of even length if it is bipartite

- **False:** If a graph is **bipartite**, the graph can be colored using two colors such that no two adjacent vertices share the same color. A bipartite graph can have cycles of **even length** (for example, a cycle of length 4). Bipartite graphs cannot have **odd-length cycles**.

E) A graph G can be colored with $O(\log|V|)$ colors if it has $O(|V|)$ edges

- **False:** It is true that sparse graphs (with $O(|V|)$ edges) tend to require fewer colors, the relationship colors are not guaranteed for general graphs with $O(|V|)$ edges.

Option C is correct.

Correct Statements:

(A) Any tree is 2-colorable

(C) A graph G is 2-colorable if it is bipartite.

(D) A graph G can be colored with $d+1$ colors if d is the maximum degree of any vertex in the graph G.

Q54 Text Solution:

Statement A

"A hash function takes a message of arbitrary length and generates a fixed-length code."

This statement is true as A hash function is designed to accept input messages of any length. Whether the message is a single character or an entire book, the hash function can process it. Despite the varying input lengths, a hash function always produces an output (also known as a hash value or digest) of a fixed size. For example:

- MD5 Hash Function: Always produces a 128-bit hash value.
- SHA-256 Hash Function: Always produces a 256-bit hash value.

This property is crucial for tasks like data integrity verification and cryptography, where consistent output sizes are needed for comparison and further processing. Hence, Statement A accurately describes how hash functions work.

Statement C

"A hash function may give the same hash value for distinct messages."

This statement is true since A hash collision occurs when two different input messages produce the same hash output.

Q55 Text Solution:

The table below outlines the decidability of various problems for different types of languages. Let's interpret the relevant rows for context-free languages (CFLs), which correspond to Pushdown Automata (PDA) for CFLs in the table.

Sr. No.	Statement	FA Regular	PDA DCFL	PDA CFL	LBA CSL	Recursive HTM	REL TM
1	Membership $W \in L$	Decidable	Decidable	Decidable	Decidable	Decidable	Undecidable
2	Finiteness $L = \text{finite}$	Decidable	Decidable	Decidable	Undecidable	Undecidable	Undecidable



3	Equiv alenc e $L_1=L_2$	Deci dabl e	Deci dabl e	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le
4	Is $L_1 \subseteq L_2$ Sub set Chec king	Deci dabl e	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le
5	Empti ness $L_1=\Phi$	Deci dabl e	Deci dabl e	Deci dabl e	Unde cidab le	Unde cidab le	Unde cidab le
6	Disjoi nt Oper ator $L_1 \cap L_2 = \Phi$	Deci dabl e	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le
7	Is $L = \Sigma^*$ Totali ty	Deci dabl e	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le
8	Is L regul ar langu age	Deci dabl e	Deci dabl e	Unde cidab le	Unde cidab le	Unde cidab le	Unde cidab le

According to the above table, rows 3,4, and 8 are undecidable. Hence the correct answer is (A), (B), and (C).

Q56 Text Solution:

Option D is Correct

Only the Language L_2 is Regular as $L_1 = \{a^{zz} \mid z \text{ is an integer}\}$

So, $L_1 = \{a^1, a^4, a^{27}, a^{256}, \dots\}$

and $L_2 = \{a^{zz} \mid z \geq 0\}$

So, $L_2 = \{a^1, a^{11}, a^{22}, a^{33}, \dots\}$

and $L_3 = \{\omega\omega \mid \omega \in \{a, b\}^*\}$

So, $L_3 =$ Set of all Strings Starting and ending with the same character.

Since L_1 does not contain any pattern to form a loop. Hence, it's not possible to construct a DFA for L_1 . On the other hand, L_2 is a regular language since a DFA can be constructed using a loop that after every 11 Symbols, the constructed Automata can be reached at Final State. In L_3 , once the word ' ω ' is scanned, there is no way to compare the next word with the starting symbol of the first word.

Q57 Text Solution:

Option A is Correct

Analysis of Grammar (A):

The grammar is defined as $S \rightarrow SS \mid aSb \mid bSa \mid \lambda$.

Since, For the empty string λ , there is only one derivation ($S \rightarrow \lambda$), so no ambiguity arises for this case. For other strings, such as abab, the derivation can occur in multiple ways due to the recursive nature of $S \rightarrow SS$. For example:

$S \rightarrow aSb \rightarrow abSa \rightarrow abab$

$S \rightarrow bSa \rightarrow baSb \rightarrow abab$

These derivations correspond to different parse trees. Therefore, Grammar (A) is ambiguous.

Analysis of Grammar (B):

The grammar is defined as $S \rightarrow asbs \mid bsas \mid \lambda$.

For the empty string λ , there is only one derivation ($S \rightarrow \lambda$), making this case unambiguous.

For strings like abab, there is only one possible derivation ($S \rightarrow asbs \rightarrow abab$). There are no alternative derivations or parse trees for any valid string. Thus, Grammar (B) is unambiguous.

Analysis of Grammar (C):

The grammar is defined as

$S \rightarrow aAB, A \rightarrow bBb, B \rightarrow A \mid \lambda$.

For strings like abbba, multiple derivations are possible due to the recursive rule $B \rightarrow A$. For instance:

$S \rightarrow aAB \rightarrow abBbB \rightarrow abbba$

$S \rightarrow aAB \rightarrow abBb \rightarrow abbba$

These derivations correspond to distinct parse trees, making Grammar (C) ambiguous.

From the analysis above, Grammar (A) and Grammar (C) are ambiguous because they allow multiple distinct parse trees for certain strings.



Grammar (B), however, is unambiguous as it permits only one parse tree for every valid string.

Q58 Text Solution:

Option A is Correct

Cloud computing relies on the pooling and sharing of resources to provide scalable and efficient services. Let's analyze each concept:

(A) Virtual Memory:

Virtual memory is a technique used in operating systems to extend physical memory by using disk space. While it's resource-efficient for local machines, it is not directly related to cloud resource pooling and sharing. Hence, It is Not related to the concept.

(B) Service:

"Service" in cloud computing (e.g., SaaS, PaaS, IaaS) represents a deliverable over the internet, but it doesn't inherently manage resource pooling or sharing. Hence, It is Not directly related to the concept.

(C) Virtualization:

Virtualization is the key enabler of cloud computing, allowing multiple virtual resources (e.g., VMs) to share the same physical hardware. This is directly related to pooling and sharing of resources. Hence, it is Directly Related to the Concept.

Only Statement C is Directly Related.

Q59 Text Solution:

Option D is Correct

Post Office Protocol (POP) is a simple message access protocol used by email clients to retrieve messages from a mail server.

(A) POP has two modes, Delete mode and Keep mode.

This Statement is True as POP provides two main modes of operation: Delete mode (where emails are removed after retrieval) and Keep mode (where emails remain on the server after retrieval).

(B) In Delete mode, mail is deleted from the mailbox after each retrieval.

This Statement is True as In Delete mode, the server removes emails from the mailbox after

they are successfully retrieved by the client.

(C) In Delete mode, mail is deleted from the mailbox before each retrieval.

This Statement is False as Emails are not deleted before retrieval in Delete mode; they are deleted after retrieval.

(D) In Keep mode, mail is deleted before retrieval.

This Statement is False as In Keep mode, emails are not deleted before retrieval; they remain on the server after retrieval.

(E) In Keep mode, mail remains in the mailbox after retrieval.

This Statement is True as The Keep mode ensures that emails are left intact on the server even after retrieval.

Q60 Text Solution:

Option D is Correct

Since, A firewall is a network security device that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Firewalls can filter packets based on various criteria:

1. Source IP Address (A):

Firewalls can block or allow packets originating from specific source IP addresses.

2. Destination IP Address (B):

Firewalls can block or allow packets heading to specific destination IP addresses.

3. TCP Source Port (C):

Firewalls can filter packets based on the TCP source port, which indicates the application on the source device sending the data.

4. UDP Source Port (D):

Firewalls can filter packets based on the UDP source port, which indicates the application on the source device sending the data.

5. TCP Destination Port (E):

Firewalls can filter packets based on the TCP destination port, which indicates the application on the destination device.

Q61 Text Solution:

Option D is Correct

The problem requires identifying which logical assertions correctly represent the statement:



"A person who is radical (R) is electable (E) if he/she is conservative (C), but otherwise is not electable."

So, Logical Interpretation of the Statement:

1. If a person is radical (R), then electability (E) depends on conservativeness (C): $R \Rightarrow (E \Leftrightarrow C)$
2. If a person is not conservative ($\neg C$), then they are not electable ($\neg E$): $R \wedge \neg C \Rightarrow \neg E$

Analyzing each Statement given,

(A) $(R \wedge E) \Leftrightarrow C$

This Statement is False as this statement asserts that being radical and electable implies conservativeness, and vice versa. However, this does not capture the full conditional relationship $R \Rightarrow (E \Leftrightarrow C)$.

(B) $R \Rightarrow (E \Leftrightarrow C)$

This Statement is True as this statement directly represents the given statement: If a person is radical, electability (E) depends on conservativeness (C).

(C) $R \Rightarrow ((C \Rightarrow E) \vee \neg E)$

This statement is False as this Statement complicates the relationship unnecessarily by adding a disjunction $\vee \neg E$, which is not implied by the statement.

(D) $(\neg R \vee \neg E \vee C) \wedge (\neg R \vee \neg C \vee E)$

This statement is True as this statement is the Conjunctive Normal Form of the statement $R \Rightarrow (E \Leftrightarrow C)$, and it correctly represents the logic.

Q62 Text Solution:

Option D is Correct

Examining all Statements,

(A) A sentence α entails another sentence β if β is true in few worlds where α is true.

It is False as Entailment means that β must be true in all worlds where α is true, not just a few.

(B) Forward chaining and backward chaining are very natural reasoning algorithms for knowledge bases in Horn form.

It is True as Both forward and backward chaining are efficient and natural for reasoning with Horn clauses, as these algorithms work well with the structure of Horn logic.

(C) Sound inference algorithms derive all sentences that are entailed.

It is False as Sound algorithms ensure that derived sentences are correct (true in all models of the knowledge base), but they do not necessarily derive all sentences that are entailed. That property is completeness, not soundness.

(D) Propositional logic does not scale to environments of unbounded size.

It is True as Propositional logic is not expressive enough for environments of unbounded size because it lacks quantifiers, making it unsuitable for such environments.

Q63 Text Solution:

Option C is Correct

Examining each Statement we get,

(A) Minimax search is breadth-first; it processes all the nodes at a level before moving to a node in the next level.

It is False as Minimax search is depth-first because it recursively explores child nodes until reaching terminal states, then backtracks. It does not process all nodes at one level before proceeding.

(B) The effectiveness of the alpha-beta pruning is highly dependent on the order in which the states are examined.

It is True as Alpha-beta pruning is more effective when better moves (closer to optimal) are examined earlier, as it allows more pruning.

(C) The alpha-beta search algorithm computes the same optimal moves as minimax algorithm.

It is True as Alpha-beta search is an optimization of minimax, providing the same results with potentially fewer node evaluations.

(D) Optimal play in games of imperfect information does not require reasoning about the current and future belief states of each player. It is False as In games of imperfect information, reasoning about belief states (what the other players know or believe) is critical for optimal play.

Q64 Text Solution:

Option D is Correct



List I	List II	Correct Match
(A) $R1 \cup R2$	$\{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$	I
(B) $R1 - R2$	$\{(2,2), (3,3)\}$	IV
(C) $R1 \cap R2$	$\{(1,1)\}$	II
(D) $R2 - R1$	$\{(1,2), (1,3), (1,4)\}$	III

Q65 Text Solution:

Option C is Correct

List I	List II	Explanation
(A) Handshaking	(II) Requires two control signals working in opposite directions	Handshaking uses two control signals for synchronization between devices.
(B) Programmed I/O	(IV) Requires CPU to check the I/O flag & perform transfer	In programmed I/O, the CPU continuously polls the I/O device to check flags.
(C) Interrupt-initiated I/O	(I) I/O interface informs the CPU that device is ready for transfer	In interrupt-initiated I/O, the device signals the CPU via an interrupt.
(D) I/O Processor	(III) Has local memory & controls a large set of I/O devices	I/O processors manage I/O operations independently with local memory and control.

Q66 Text Solution:

Option D is Correct

List I (Matrix)	List II (Transformation Diagram)	Explanation
(A) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(II) Scale (2, 2)	This matrix represents scaling by a factor of 2 in both x and y directions.
(B) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	(IV) Translate (2, 2)	This matrix represents translation by vector (2,2) in x and y directions.
(C) $\begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(III) Rotate (60°)	This matrix represents a counterclockwise rotation of 60°.
(D) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(I) Shear-x (2)	This matrix represents a shear transformation with a factor of 2 along the x-axis.

Q67 Text Solution:

Option C is Correct

List I (E-R symbols)	List II (Description)	Correct Match	Explanation
(A) Rectangle	(II) Weak Entity Type	A-II	Rectangles represent weak entities in an E-R diagram.
(B) Oval with	(I) Key Attribute Type	B-I	Key Attribute



List I (E-R symbols)	List II (Description)	Correct Match	Explanation
Underline Text			
(C) Double Ellipse	(IV) Multivalued Attribute Type	C-IV	Double Ellipse Represents Multivalued Attribute in ER Diagram
(D) Total Participation (double line)	(III) Total Participation of Entity in a relation	D-III	Double lines indicate total participation of an entity in a relationship.

Q68 Text Solution:

Option C is Correct

List I	List II	Explanation
(A) INITIAL	(IV) ADHOC	The initial level is unstructured and chaotic, representing ad-hoc processes.
(B) REPEATABLE	(III) Project Management	At this level, processes are repeatable and rely on basic project management.
(C) DEFINED	(II) Process Definition	At this level, processes are clearly defined, standardized, and documented.
(D) MANAGED	(I) Process Measurement	The managed level focuses on measurable and controlled processes.

Q69 Text Solution:

Option A is Correct

List I	List II	Explanation
(A) Branch-and-bound	(I) Keeps track of all partial paths which can be candidates for further exploration	Branch-and-bound explores all possible solutions while maintaining the best candidate paths.

List I	List II	Explanation
(B) Steepest-descent hill climbing	(IV) Considers all moves from current state and selects the best move	Hill climbing is a greedy algorithm that selects the steepest or best move at each step.
(C) Constraint satisfaction	(III) Discovers problem state(s) that satisfy a set of constraints	Constraint satisfaction focuses on finding states that meet a set of given conditions.
(D) Means-end-analysis	(II) Detects difference between current state and goal state	Means-end-analysis involves reducing the difference between current and goal states iteratively.

Q70 Text Solution:

Option C is Correct

List I	List II	Explanation
(A) Topological sort of DAG	$\Theta(V+E)$ (III)	Processes vertices and edges once.
(B) Kruskal's MST Algorithm	$O(V+E)$ (I)	Assumes edges are pre-sorted or sorted in linear time.
(C) Bellman-Ford's Single-Source Shortest Path Algorithm	$O(V^2E)$ (II)	Iterates $V-1$ times over E edges.
(D) Floyd-Warshall's All-Pairs Shortest Path Algorithm	$\Theta(V^3)$ (IV)	Uses triple nested loops for all-pairs processing.

Q71 Text Solution:

Option C is Correct

List I	List II	Explanation
(A) Recursively Enumerable	$\overline{L}_{REC} \cup L_{RE}$ (I)	The union of recursive and recursively enumerable



List I	List II	Explanation
able Language		languages is recursively enumerable.
(B) Recursive Language	$\overline{L_{CF}} \cup L_{REC}$ (II)	Recursive languages include the complement of context-free languages; their union remains recursive.
(C) Context-Free Language	$L_R^* \cup L_{CF}$ (III)	The intersection of regular and context-free languages results in a context-free language.

Q72 Text Solution:

Option B is Correct

List I	List II	Explanation
(A) Serial Line IP (SLIP)	Data Link Layer (III)	SLIP operates at the Data Link layer for encapsulating IP packets over serial links.
(B) Border Gateway Protocol (BGP)	Network Layer (IV)	BGP is used for routing between autonomous systems and operates at the Network Layer.
(C) User Data Protocol (UDP)	Transport Layer (II)	UDP is a Transport Layer protocol that provides connectionless communication.
(D) Simple Network Management Protocol (SNMP)	Application Layer (I)	SNMP operates at the Application Layer to manage network devices.

Q73 Text Solution:

Correct option is A.

Greedy best-first search algorithm always selects the path which appears best at that moment. It is the combination of depth first search and breadth first search algorithms.

Time complexity - The worst case time complexity of the greedy best first search is $O(b^m)$.

Space complexity - The worst case space complexity of the greedy best first search is $O(b^m)$. Where m is the maximum depth of the search space.

Complete - Greedy best first search is also incomplete, even if the state space is finite.

Optimal - Greedy best first search algorithm is not optimal.

-> A^* is optimal and complete but time complexity and space complexity too high. It requires more computation.

-> Recursive Best first Search retains more memory knowledge but uses only $O(bd)$ memory.

-> Memory Simplified Bound A^* such as A^* Optimal if you can get optimal solution otherwise as it is memory bound.

Q74 Text Solution:

Correct option is B.

Lexicographic order is nothing but dictionary order.

To determine the order of strings, we need to compare each bit of all the strings one by one. For example, take 001 and 010. Here, the second bits of the strings are 0, 1 respectively.

Since $0 < 1$ as per the given question, $001 < 010$.

Similarly, $0001 < 001 < 010 < 0101 < 011$

Q75 Text Solution:

Correct option is C.

MIMD (Multiple instruction stream, multiple data stream)- This includes machines with multiple processing units that can apply multiple instructions to different data at the same time.

The MIMD machines are the most complicated, but they also hold the most promise for increasing productivity by concurrent processing. Concurrency means here that not only are multiple processors running at the same time but also that multiple programs are running at the same time, concurrently.



SIMD - SIMD is an abbreviation for Single Instruction and Multiple Data Stream. It symbolises an organization with a large number of processing units overseen by a common control unit. The control unit sends the same instruction to all processors, but they work on separate data.

SISD - SISD is a single-processor machine that is capable of executing a single instruction. It is a classification of computer architectures, and it operates on just a single data stream.

Q76 Text Solution:

Correct option is C.

3D projection	Properties
(1)a perspective projection only	Requires homogeneous coordinates in order for it to be encoded into a linear transformation
(2)an orthographic projection only	Far away objects appear the same size as closer ones (as orthographic projection preserve dimensions)
(3)both orthographic and projective	Straight lines are mapped to straight lines
(4)neither orthographic nor projective	Distances and angles are (in general) preserved

Q77 Text Solution:

The correct option is D.

A design solution is said to be highly modular if the different modules in the solution have high cohesion and their inter-module coupling is low. Cohesion is like a type of ranking which is used to measure the degree of modules that are functionally related.

The degree of Cohesion can be defined as from best to worst-

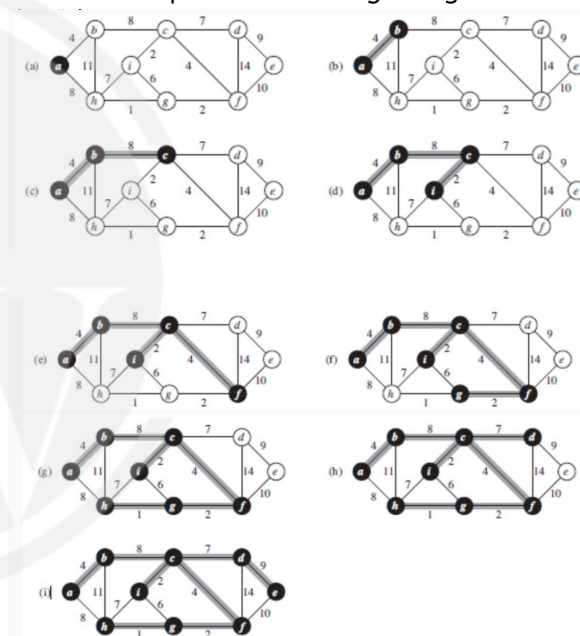
Functional
Sequential

Communicational
Procedural
Temporal
Logical
Coincidental

Q78 Text Solution:

The correct option is C.

A minimum spanning tree (MST) is a subset of the edges $(V-1)$ of a connected, edge-weighted undirected graph $G(V, E)$ that connects all the vertices together, without any cycles and with the minimum possible total edge weight.



So the correct option will be C according to the above diagram.

Q79 Text Solution:

Correct option is B.

$L(r) = \{ a, aa, ab, aaa, aab, \dots \}$ This is a set of all strings starting with 'a'.

$L(s) = \{ aa, ab, aaa, aab, \dots \}$ This is a set of all strings starting with 'aa' or 'ab'.

$L(t) = \{ ab, aab, aaab, aaaab, \dots \}$ This is a set of all strings starting with 'a', ending with 'b' and contains any number of 'a's in between them.



Therefore, $L(r) \supseteq L(s) \supseteq L(t)$

Q80 Text Solution:

Correct option is A.

Statement 1: 5 divides $n^5 - n$

Using mathematical Induction, if $n^5 - n$ is divisible by 5, then $(n + 1)^5 - (n+1)$ is also divisible by 5.

if 5 divides, $n^5 - n = 5k$

$$(n + 1)^5 - (n+1) = n^5 + 5n^4 + 10n^3 + 10n^2 + 5n - n = 5k + 5(n^4 + 2n^3 + 2n^2 + n)$$

$$\text{This shows that } (n + 1)^5 - (n+1) = n^5 + 5n^4 + 10n^3 + 10n^2 + 5n - n$$

$$= 5k + 5(n^4 + 2n^3 + 2n^2 + n)$$

This shows that $(n + 1)^5 - (n+1)$ is a multiple of 5.

Hence it is always divisible by 5.

Statement 2: 6 divides $n^3 - n$

$$n^3 - n = n(n^2 - 1) = n(n+1)(n-1)$$

Here n , $n+1$ and $n-1$ are three consecutive integers. One of them is a multiple of 2 and the other is a multiple of 3. Hence, 6 divides $n^3 - n$ is always divisible by 6 whenever n is a non-negative integer

Q81 Text Solution:

Correct option is A.

Control signals are provided by a hardwired circuit in hardwired control techniques. The main goal of the control unit is to produce the control signal in the correct order. The speed of a hardwired control unit can be optimized. So statement I is correct.

Indirect addressing mode needs two memory references to fetch operand as the address of the operand is not given in instruction but the address field contains the address where the address of the operand is stored. So Statement II is also correct.

Q82 Text Solution:

Correct option is D.

The Bezier curve is defined by P_0 through P_n and n is named its order ($n=1$ for the linear, 2 for the quadratic, etc.). The first and last checkpoints are often the endpoints of the curve, but interim checkpoints do not usually lie on the curve. So Statement 1 is incorrect.

A cubic bezier curve has four control points. So the statement 2 is correct.

Q83 Text Solution:

Correct option is C.

Both data points would be in a straight line if the correlation between the variables V_1 and V_2 is 1. So, all three cluster centroids will form a straight line.

Q84 Text Solution:

The correct option is C.

`fork()` is a system call used to create a child process.

Once a child process is created, both the parent process and the child process run the next line following the `fork()` command simultaneously. 'n' number of successive `fork()` calls create $2^n - 1$ child processes.

in P1: Each child process prints one "Happy".

Parent process also prints one "Happy".

So, number of times P1 displays "Happy" = $(2^3 - 1) + 1 = 8$

In P2: At first `fork()`, a new child process with two `fork()` and `printf` statements is created.

More child processes are created at each `fork()` command.

So, p2 shows "Happy" 14 $(2+4+8)$ times.

Q85 Text Solution:

Correct option is A.

Quality control is a series of inspections, reviews, and tests conducted during the software development process to ensure that each work product follows the specifications. A feedback loop to the process that produced the work product is included in quality control.

So Statement 1 is true.

Management's auditing and monitoring activities make up quality assurance. The aim of quality assurance is to provide management with the information they need to make informed decisions about product quality, allowing them to gain knowledge and trust that the product is achieving its objectives.

So Statement 2 is also correct.



Q86 Text Solution:

There are four types of edges that can yield in DFS. These are tree, forward, back, and cross edges. In an undirected connected graph, the forward and back edges are the same thing. A cross edge in a graph is an edge that goes from a vertex v to another vertex u such that u is neither an ancestor nor descendant of v . Therefore, a cross edge is not possible in an undirected graph. So, statement (I) is correct. For statement (II) take a counterexample of the complete graph of three vertices, i.e., K_3 with XYZ, where X is the source and Y and Z are at the same level. Also, there is an edge between vertices Y and Z, i.e., $|i-j| = 0 \neq 1$ in BFS. So, the statement became false.

Q87 Text Solution:

-> The problem is " $L_1 \wedge L_2 = \emptyset$ " is undecidable for context sensitive language L_1 and L_2 .

-> The problem is $W \in L$ is decidable for context sensitive language. Here W is a string that is accepted by the particular language is a membership problem and is a decidable problem.

Q88 Text Solution:

Explanation:

I. The laws of nature put two fundamental limits on the data rate of a channel.

The H.Nyquist limit restricts the number of independent samples per second to twice the bandwidth in a Noiseless channel. ($R=2B$)

II. Highest data rate for a noisy channel:

Capacity $= H \times \log_2 \left(1 + \frac{S}{N}\right)$ Where H is bandwidth

H = Bandwidth of the channel in Hertz.

S = Average signal power.

N = Average noise power.

The Final Answer is: Both Statement I and Statement II are true

Q89 Text Solution:

Explanation:

The Caesar Cipher is a basic encryption method where each letter in the plaintext is substituted with a letter that is a fixed number of positions forward in the alphabet.

(Example: Applying a shift of 3 to the word PQR would result in the encrypted text STU)

Diffie-Hellman enables the creation of a shared secret without exposing the key during transmission, ensuring secure key exchange.

The Final Answer is: Both Statement I and Statement II are true

Q90 Text Solution:

Explanation:

A genetic algorithm is a stochastic hill-climbing search in which a large population of states is maintained.

AND-OR search is used to generate contingent plans in nondeterministic environments.

The Final Answer is: Both Statement I and Statement II are true

Q91. Text Solution:

Explanation:

Sector size = 512 bytes

Sectors per track = 50 sectors

Tracks per surface = 2000 tracks per surface

Capacity of track = Number of Sectors per track

$\times$ Sector size = 50 $\times$ 512 bytes = 25600 bytes = 25K

Capacity of each surface = capacity of track $\times$

Number of tracks per surface = 25K $\times$ 2000 = 50000K

The Final Answer is: (25 K, 25000K)

Q92. Text Solution:

Explanation:

Capacity of each surface = capacity of track $\times$

Number of tracks per surface = 25K $\times$ 2000 = 50000K

Total Disk Capacity = Capacity of one surface $\times$

Number of surfaces = 50,000KB $\times$ 10 = 500,000KB

(Number of surfaces = 5 double-sided platters and each platter has two surfaces = 5 $\times$ 2 = 10).

The Final Answer is 500,000 K.

Q93. Text Solution:

Explanation:

Correct: The number of cylinders is 2000 equal to a number of tracks on each platter.

Correct: The block size will be a multiple of the sector size. 51200 is not a valid block size in this case because block size cannot exceed the size of a track, which is 25600 bytes.

The Final Answer is: A

Q94. Text Solution:

Explanation:

if the disk platters rotate at 5400rpm, the time required for one complete rotation is the maximum rotational delay.

Therefore, 5400 rotations= 1 minute

⇒ one rotation = $\frac{1}{5400}$ minute

⇒ one rotation = $\frac{1}{5400}$ X 60 seconds

⇒ maximum rotational delay = 0.011seconds

The Final Answer is: 0.011 seconds

Q95. Text Solution:

Explanation:

Capacity of track = Number of Sectors per track

X Sector size = 50 X 512 bytes = 25600 bytes= 25K

Rotation delay = 0.011 seconds

Now, the Data transfer rate = $\frac{\text{capacity of track}}{\text{rotation delay}} =$

$\frac{25 \text{ K}}{0.011 \text{ seconds}} = \frac{25600 \text{ Bytes}}{0.011 \text{ seconds}} \approx 2250$

KBytes/second

The Final Answer is: 2,250 KBytes/second

Q96. Text Solution:

Explanation:

SELECT DISTINCT EMPID

FROM PROJECTWORK

WHERE PROJNO IN (20,30,40);

Primary Key (EMPID, PROJNO) same EMPID may be repeated for a PROJNO in keyword is required.

Where a **distinct** keyword is needed and to match with any of multiple values os PROJNO IN keyword is required.

The Final Answer is: C

Q97. Text Solution:

Explanation:

Using the Join condition(two tables) must be matched so that records can be linked uniquely. SUM (SALARY). MAX (SALARY), MIN (SALARY), AVG (SALARY) FROM EMPLOYEE, DEPARTMENT WHERE DEPTNO DID AND DNAME='ENGLISH': Without DEPTNO DID condition as in statement II, there will be the cartesian product of two tables (EMPLOYEE, DEPARTMENT) and it will appear many duplicate rows with DNAME='ENGLISH'.

(The aggregation functions calculate values based on these duplicates, producing incorrect results.)

The Final Answer is: Statement I is correct but Statement II is false

Q98. Text Solution:

Explanation:

SELECT EID, NAME

FROM EMPLOYEE

WHERE SALARY>ALL(SELECT SALARY FROM EMPLOYEE WHERE DEPTNO =20)

ORDER BY EID

The query returns the EID and name of employees whose salary is greater than the salary of all employees in department number 20 of the university. Order results by employee ID.

The Final Answer is: Option D

Q99. Text Solution:

Explanation:

If we use the RESTRICT option, the column drop will be rejected if it invalidates a dependent schema object. If CASCADE is defined, the column drop should also remove any other schema objects that have become invalid. Since the salary field is not present in any other table, CASCADE and RESTRICT have no impact.

The Final Answer is: Option A

Q100. Text Solution:

Explanation:

UPDATE table_Name



SET column 1= value 1, column 2=
value 2, column 3= value 3,.....
WHERE condition;

UPDATE EMPLOYEE

SET SALARY = SALARY * 1.20

WHERE DEPT NO. IN (SELECT DID FROM
DEPARTMENT WHERE DNAME='SCIENCE');

The Final Answer is: Option B



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